3
Natural Theology

Aquinas famously tells us in his *Summa Theologiae* that "the existence of God can be proved in five ways" (*ST* I.2.3). They are (in the order in which he there presents them) the *proof from motion*, the *proof from causality*, the *proof from the contingency of the world*, the *proof from the grades of perfection*, and the *proof from finality*. The short passage in which he states these proofs has appeared in countless anthologies aimed at undergraduates and general readers, and it may be the most famous set of arguments for God's existence ever written. No doubt many readers take the Five Ways to be Aquinas's complete case for the existence of God, indeed, *the* complete case for the existence of God, full stop (apart perhaps from St. Anselm's famous ontological argument). Hence, those who read them and remain unconvinced may conclude from that fact alone that the case for God's existence simply hasn't been made, by Aquinas or likely anyone else.

This is unfortunate, and certainly unfair. To be sure, Aquinas is probably the greatest philosopher of religion in the Western tradition, and though many other thinkers have presented interesting and influential arguments for God's existence, it is not unreasonable to regard Aquinas's work as representative. Moreover, he did think that the best arguments that could be given for God's existence are summarized in the Five Ways. (He rejected Anselm's ontological argument, for reasons we will see later.) But it is crucial to understand that they are *summaries*. Aquinas never intended for them to stand alone, and would probably have reacted with horror if told that future generations of students would be studying them in isolation, removed from their original immediate context in the *Summa Theologiae* and the larger context of his work as a whole. The *Summa*, it must be remembered, was meant as a textbook for beginners in theology who were already Christian believers, not an advanced work in apologetics intended to convince skeptics. The Five Ways themselves are merely short statements of arguments that would already have been well known to the readers of Aquinas's day, and presented at greater length and with greater

precision elsewhere. For example, he gives two much more detailed versions of the proof from motion, along with versions of the proofs from causality, the grades of perfection, and finality, in the *Summa contra Gentiles*. The proof from motion, having originated with Aristotle, is also naturally discussed at length in Aquinas's commentaries on Aristotle's *Physics* and *Metaphysics*. The *Commentary on the Sentences*, *On Being and Essence*, *On Truth*, and the *Compendium of Theology* each contain further statements of some of the arguments. Some of them were also familiar from the works of Christian thinkers like St. Augustine, St. John Damascene, and Albert the Great, Muslim thinkers like Avicenna and Averroes, and the Jewish philosopher Moses Maimonides. That the being whose existence Aquinas takes the Five Ways to have proved must have all the divine attributes is something he devotes much of the rest of Part I of the *Summa Theologiae* (as well as hundreds of pages of his other works) to proving. And of course the metaphysical ideas apart from which the Five Ways cannot properly be understood (and which were surveyed in the previous chapter) are developed throughout Aquinas's works.

Torn from this rich context, as they so frequently are, it is no surprise that the Five Ways have been regarded by some readers as anticlimactic or worse. For instance, in his atheistic polemic *The God Delusion*, Richard Dawkins asserts boldly that the arguments "don't prove anything, and are easily – though I hesitate to say so, given [Aquinas's] eminence – exposed as vacuous." But Dawkins' confidence is misplaced, for the objections he makes are based on egregious misunderstandings of the Five Ways of the sort that are bound to arise when one reads only a short anthologized selection from the *Summa* and ignores the metaphysical concepts which underlie the arguments. Dawkins claims, for example, that Aquinas holds that since "there must have been a time when no physical things existed," something must have brought them into being. But in fact Aquinas famously thought that it cannot be proven philosophically that the world had a beginning in time, and while he nevertheless believed it did, he held that this was something that could be known only through divine revelation (*ST* I.46.2). Consequently, his arguments are *not* intended to show that God caused the world to begin at some point in the past (at the Big Bang, say). Rather, he argues that even if the world had always existed, God would still have to exist *here and now*, otherwise certain features that it exhibits *here and now* would be inexplicable. Dawkins also alleges that Aquinas gives "absolutely no reason" to think that the cause of the world must be omnipotent, omniscient, good, and so on. In fact, and as noted already, Aquinas devotes a great many pages to showing this, as anyone who takes the trouble to read the *Summa Theologiae* beyond the passage containing the Five Ways will soon discover. Dawkins

thinks that Aquinas's Fifth Way is more or less the same as William Paley's famous "argument from design," when in fact they are radically different, since Aquinas's argument appeals to Aristotelian teleology while Paley's assumes instead a non-teleological mechanistic conception of the natural world. And so forth.

Other common objections to the Five Ways are based on similar misunderstandings. For example, the Second Way is often thought to say that since everything has a cause, the universe too must have a cause, which is what we call God. It is then objected that the argument undermines itself, since if "everything has a cause," then this would have to include God too, in which case he cannot be the first cause. But that is not what the proof says at all. Aquinas does not hold that "*everything* has a cause." He holds instead only that *that which comes into being*, and more generally *that which is contingent*, must have a cause. (This, you will recall from chapter 2, is the Thomistic "principle of causality.") Obviously there is nothing in this that entails that God would have to have a cause, since God is supposed to have always existed as a necessary being. Whether one thinks the Second Way ultimately works or not, it does not commit the simple and obvious fallacy of which popular accounts of the argument sometimes accuse it.

It has also sometimes been claimed (for example by Anthony Kenny) that Aquinas's proofs rest on outdated Aristotelian scientific theory, and thus are irrelevant in the present day. But as noted in chapter 2, Aristotle's metaphysics stands or falls independently of his physics, and as we shall see, while the Five Ways definitely presuppose certain Aristotelian metaphysical claims, there is never a point in any of the arguments where appeal need be made to now falsified theories in physics or any of the other sciences. Indeed, we will see that the Five Ways remain as interesting and worthy of consideration today as any other philosophical argument.

The First Way

As presented in the *Summa Theologiae*, the proof from motion goes as follows. We know from experience that "some things are in motion" ("motion" in the Aristotelian sense just being change, as we saw in our discussion of Aristotle's reply to Parmenides). Now motion or change is just the reduction of something from potentiality to actuality. But "nothing can be reduced from potentiality to actuality except by something in a state of actuality" (*ST* I.2.3); for instance, fire, which is actually hot, makes wood, which is otherwise only potentially hot,

become actually hot. Moreover, nothing can be both potential and actual in the same respect at the same time; what is actually hot, for example, is not at the same time potentially hot, but potentially cold. In that case, though, it is impossible for anything to be at the same time and in the same respect both that which is moved or changed and that which does the moving or changing. Hence, "whatever is in motion must be put in motion by another" (*ST* I.2.3). By the same token, if that which puts something else in motion is itself moving, there must be yet something further moving *it*, and so on. But if such a series went on to infinity, then there would be no first mover; and if there were no first mover, there would be no other movers, for "subsequent movers move only inasmuch as they are put in motion by the first mover; as the staff moves only because it is put in motion by the hand" (*ST* I.2.3). It follows that "it is necessary to arrive at a first mover, put in motion by no other; and this everyone understands to be God" (*ST* I.2.3).

To begin at the end, someone might immediately object to this argument that whatever else Aquinas has shown, he hasn't really shown that such a "first mover" would be God, if by God we mean a being that can be said to be all powerful, all knowing, all good, and the like. There are two things to be said in reply. First, what Aquinas is getting at in the last line of the proof is that whatever else God is supposed to be, he is supposed to be the ultimate explanation of why things happen in the world; hence, if it can be proved that there is a being who explains this, it follows that at least to that extent it will have been proved that there is something in reality corresponding to our idea of God. And he is surely right about that much. Second, while we do of course also want to know why we should regard such a being as all-powerful, all-knowing, all-good, and so forth, as I have said before, Aquinas does in fact answer that question in great detail later on in the *Summa* (and elsewhere). We will see how he does so after first looking at each of the Five Ways.

The question for now, then, is this: does this argument really establish the existence of a first Unmoved Mover? Note first of all that the argument cannot be criticized by appealing to a variation on the standard "If everything has a cause, then what caused God?" objection. Aquinas does not say that everything is in motion, but only that "some things" are in motion; nor does he say that everything is moved by something else, but only that "whatever is in motion" is moved by something else. Hence it will not do to ask "Doesn't that mean that God must be in motion?" or "What moves God, then?" For there is nothing in Aquinas's premises that implies that God would have to be changing like everything else is, or that he must be moved by something else.

What, then, of this key premise of the argument, that is, that "whatever is in

motion must be put in motion by another"? The bulk of the proof is devoted to supporting it. Yet it has often been suggested that Aquinas's argument for it fails. One common objection is that the activity of animals shows that the premise is simply false. For isn't it just obvious that animals move themselves? But as we noted in chapter 2, Aquinas does not deny that there is a loose sense in which animals move themselves. Strictly speaking, though, when an animal moves this only occurs because one part of the animal moves another part, as when the legs of a dog move because of the flexing of its muscles, the muscles flex only because of the firing of certain motor neurons, and so forth. When considered in detail, then, the example of animal movement does not constitute a counterexample to the principle that "whatever is moved is moved by another."

It is also sometimes alleged that Aquinas is committed to the principle that whatever causes something actually to be *F* must itself actually be *F*, and that this principle is clearly false. For he gives the example of wood being made to catch fire by something which is already on fire; but as Kenny points out, fire could also be generated instead by taking two sticks that are not already on fire, and rubbing them together. But there are two problems with this objection. First, it ignores the possibility that Aquinas is here appealing to what we called in chapter 2 the "principle of proportionate causality," according to which whatever is in an effect must somehow be in its cause, but where this allows that the cause might have the relevant feature "virtually" or "eminently" rather than "formally." In other words, Aquinas is not making the obviously false claim that only what is already on fire can cause fire; he is rather making the claim (perfectly defensible, as we saw in the previous chapter) that whatever causes fire must have an inherent power to cause it. Second, as many commentators have pointed out, Aquinas is probably not relying in this argument on any version of the principle in question in the first place. That is, he is not saying that "whatever causes something actually to be *F* must itself be *F* in some way," but rather that "whatever causes something must itself be actual," that nothing merely potential can cause anything. As Rudi te Velde has suggested, some critics place too much significance on the physical details of the examples Aquinas gives in the course of the proof, failing to see that their point is merely to illustrate certain basic metaphysical principles rather than to support broad empirical or quasi-scientific generalizations.

Thus understood, what Aquinas is saying here is essentially just what we have already noted him saying in developing the distinction between act and potency, namely that no potency or potential can actualize itself, precisely because it is merely potential and not actual. Hence only what is itself already actual can actualize a given potency, and therefore (given that motion is just the

actualization of a potency) “whatever is moved is moved by another.” This is not some dubious conjecture based on the observation of how wood catches fire and the like; it is rather supposed to be a metaphysical certainty the denial of which would be conceptually incoherent. Indeed, the principle in question is but a variation on what we referred to in chapter 2 as the “principle of causality,” which we have seen to be eminently defensible.

So far, so good, then. But what about the claim that a series of movers could not go on to infinity? Isn’t Aquinas just begging the question (arguing in a circle) when he asserts that if there were no first mover then there would be no movers at all? For why could there not be an infinite series of movers, so that no matter how far back you go in the series, you could always go back to yet another mover? In that case it seems there would be an explanation for the motion of any member of the series you care to take, without having to appeal to a first mover.

But in fact Aquinas is not begging the question at all, and has good reason for claiming that such a series could not go on to infinity. Keep in mind first of all that the proof from motion, like all the Five Ways, is not an attempt to show that the universe had a beginning at some point in the past and that God must have caused that beginning. Aquinas is not saying that if you trace the series of movers back in time you must eventually get to some temporally first mover. As we saw in chapter 2, for Aquinas as for Aristotle, the immediate cause of an effect is *simultaneous* with that effect: “It is clear that when a thing moves because it is moved, the mover and the mobile object are moved simultaneously” (*In Phys* VII.2.892). So the series of movers he has in mind is one all of whose members exist together *here and now* (and at any moment we might be considering the argument), and by saying that there must be a first mover, he doesn’t mean first in order of time, but rather first in the sense of being most fundamental in the order of what exists.

This brings us to an important distinction Aquinas and other medieval thinkers made between two kinds of series of efficient causes. On the one hand there are causal series ordered *per accidens* or “accidentally,” in the sense that the causal activity of any particular member of the series is not essentially dependent on that of any prior member of the series. Take, for example, the series consisting of Abraham begetting Isaac, Isaac begetting Jacob, and Jacob begetting Joseph. Once he has himself been begotten by Abraham (and then grows to maturity, of course), Isaac is fully capable of begetting Jacob on his own, even if Abraham dies in the meantime. It is true that he would not have existed had Abraham not begotten him, but the point is that once Isaac exists he has the power to beget a son all by himself, and Abraham’s continued existence or non-existence is

irrelevant to his exercise of that power. The same is true of Jacob with respect to both Abraham and Isaac, and of Joseph with respect to Abraham, Isaac, and Jacob. Given that we are considering them as a series of begetters specifically, each member is independent of the others as far as its causal powers are concerned. Contrast this with a causal series ordered per se or "essentially." Aquinas's example from the First Way of the staff which is moved by the hand is a standard illustration, and we can add to the example by supposing that the staff is being used to move a stone, which is itself moving a fallen leaf. Here the motion of the leaf depends essentially on the motion of the stone, which in turn depends essentially on the motion of the staff, which itself depends essentially in turn on the motion of the hand. For if any member higher up in the series ceases its causal activity, the activity of the lower members will necessarily cease as well. For instance, if the staff was to slip away from the stone, the stone, and thus the leaf too, will stop moving; and of course, if the hand stops moving, the whole series, staff included, will automatically stop. In this case the causal power of the lower members derives entirely from that of the first member, the hand. In fact, strictly speaking it is not the stone which is moving the leaf and the staff which is moving the stone, but rather the hand which is moving everything else, with the stone being used by it as an instrument to move the leaf and the staff being used as an instrument to move both stone and leaf.

Causal series ordered *per accidens* are linear in character and extend through time, as in the begetting example, in which Abraham's begetting Isaac occurs well before Isaac's begetting Jacob, and Isaac's begetting Jacob occurs well before Jacob's begetting Joseph. Causal series ordered per se are paradigmatically hierarchical with their members acting simultaneously, as in the staff example where the movement of the leaf occurs precisely when the movement of the stone occurs, which is precisely when the movement of the staff occurs, which is precisely when the movement of the hand occurs. Now it is in Aquinas's view at least theoretically possible for a causal series ordered *per accidens* to regress to infinity, and thus have no beginning point (*ST* I.46.2). (This is why Aquinas thinks it is not possible to prove via purely philosophical arguments that the world must have had a beginning in time.) For since each member of such a series has its causal power independently of the earlier members, there is no need to trace any particular member's action back to the activity of a first member; for instance, when Jacob begets Joseph, it is precisely Jacob who begets him, and not Abraham who begets him by using Isaac and Jacob as instruments. By contrast, "in efficient causes it is impossible to proceed to infinity per se – thus, there cannot be an infinite number of causes that are per se required for a certain effect; for instance, that a stone be moved by a stick, the

stick by the hand, and so on to infinity" (*ST* I.46.2). For "that which moves as an instrumental cause cannot move unless there be a principal moving cause" (*SCG* I.13.15). That is to say, since the lower members of a causal series ordered per se have no causal power on their own but derive it entirely from a first cause, which (as it were) uses them as instruments, there is no sense to be made of such a series having no first member. If a first member who is the source of the causal power of the others did not exist, the series as a whole simply would not exist, as the movement of the leaf, stone, and staff cannot occur in the absence of the hand.

What Aquinas is saying, then, is that it is in the very nature of causal series ordered per se to have a first member, precisely because everything else in the series only counts as a member in the first place relative to the actions of a first cause. To suggest that such a series might regress infinitely, without a first member, is therefore simply unintelligible. The leaf is "moved" by the stone only in a loose sense; strictly speaking, the leaf, stone, and staff are all really being moved by the hand. Thus to suggest that this series of purely instrumental causes might regress to infinity is incoherent, for they would not in that case be the instruments *of* anything at all (*CT* I.3). As A. D. Sertillanges put it, you might as well say "that a brush can paint by itself, provided it has a very long handle" (quoted by Garrigou-Lagrange in *God: His Existence and His Nature*).

Given their essentially instrumental character, all causes in such a series other than the first cause are referred to by Aquinas as "second causes" ("second" not in the sense of coming after the first but before the third member of the series, but rather in the sense having their causal power only in a secondary or derivative way). It is worth emphasizing that it is precisely this instrumental nature of second causes, the dependence of whatever causal power they have on the causal activity of the first cause, that is the key to the notion of a causal series per se. That the members of such a series exist simultaneously, and that the series does not regress to infinity, are of secondary importance. As Patterson Brown and John Wippel point out, even if a series of causes ordered per se could somehow be said to regress to infinity, it would remain the case, given that they are merely instrumental causes, that there must then be something outside the entire infinite series that imparts to them their causal power.

Whether or not the series of causes *per accidens* regresses infinitely into the past, then – and again, while Aquinas believed that it did not, he didn't think this could be proven through philosophical arguments – a causal series per se existing here and now, and at any moment we are considering the matter, must necessarily trace back to a first member. But strictly speaking, even the hand in Aquinas's example doesn't count as a first mover – the example is intended

merely as a first approximation to the notion of a first mover – because it is itself being moved insofar as its activity depends on the motion of the arm, the flexing of certain muscles, and so forth. To understand the way in which such a series regresses and how it *does* and must terminate, it is crucial to remember that for Aquinas, motion or change is just the reduction of potency to act. So when we talk about one thing being moved by another, which is moved by another, and so on, in a causal series ordered per se, this is shorthand for saying that a certain potency is reduced to act by something whose potency is itself reduced to act by something whose potency is itself reduced to act by … and so forth. (Or, to soften the technical terminology slightly, a certain potentiality is actualized by something whose potentiality is itself actualized by something whose potentiality is itself actualized by … and so on.) As should be evident, such a series can only possibly terminate in something which is not reduced to act or actualized by anything else, but which just *is* in act or actual, and thus "unmoving." The potential of the hand for movement is actualized here and now by the flexing of the muscles of the hand, the potential of the muscles to flex is actualized here and now by the firing of certain motor neurons, the potential of the motor neurons to fire is actualized here and now by the firing of certain other neurons, and so forth. Eventually this regress must terminate in something which here and now actualizes potentialities without itself being actualized, an unmoved mover.

Now Kenny objects that the notion of an unmoved mover merely gives us something at rest, like a stationary billiard ball, and thus seems hardly relevant to proving the existence of God. But as Garrigou-Lagrange points out, and as should be clear from our discussion thus far, an unmoved mover of the sort we've been describing is not and cannot be "unmoved" in the sense of being in repose, precisely because it is that which actualizes the potencies of second causes. It is *active*, not "at rest." There is still a further question, however. Even if it is granted that the First Way takes us to an *unmoved* mover, why should we hold (as Aquinas does) that this mover is also *unmovable*? As Scott MacDonald suggests, it may be that a first mover of the sort whose existence is established by Aquinas's argument is one that is *capable* of motion even if, qua first mover, it does not *in fact* move. In other words, for all Aquinas has shown, a first mover may well *have* certain potencies which are not in fact being *actualized*, at least not insofar as it is functioning as the first mover in some series of efficient causes ordered per se. Perhaps its potencies are actualized at some other time, when it is not so functioning; or perhaps they never are. But as long as it has them, it will not be something that can be characterized as "pure act," and thus, given Aquinas's own commitments, it will not be identifiable with God. To get

to a first mover of pure act, and thus one which is truly unmovable, would require in MacDonald's view some further argument, in which case the argument from motion could succeed as an argument for God's existence only by being "parasitic" on such a further argument.

Yet MacDonald is, I think, mistaken. Consider how the series we have been describing would have to continue beyond the point at which we left it, with the hand's potentiality for motion actualized by the arm, the arm's potentiality for motion actualized by the flexing of certain muscles, the muscles' potentiality for flexing actualized by the firing of certain motor neurons, and so on and so forth, all simultaneously. All of this depends in turn on the overall state of the nervous system, which depends on its molecular structure, which depends on the atomic basis of that molecular structure, which depends on electromagnetism, gravitation, the weak and strong forces, and so on and so forth, all simultaneously, all here and now. That the molecules composing the nervous system constitute a nervous system specifically amounts to their having a certain potency which is here and now actualized, that the atoms composing the molecules constitute just those molecules amounts to their having a certain potency which is simultaneously actualized, and so on. To account for the reduction of potency to act in the case of the *operations or activities* of the hand, the muscles, and so on, we are led ultimately to appeal to the reduction of potency to act vis-à-vis the *existence or being* of ever deeper and more general features of reality; for "it is evident that anything whatever operates so far as it is a being" (*QDA* 19). But the only way to stop this regress and arrive at a first member of the series is with something whose very existence, and not merely its operations or activities, need not be actualized by anything else. This would just be something which, since it simply exists without being made to exist by anything, or is actual without being actualized, is pure act, with no admixture of potentiality whatsoever. For suppose it had some potency relevant to its existence (its existence being what is relevant to its status as the end of the regress as we have continued it). Then either some other thing actualizes that potency, in which case we haven't really stopped the regress after all, contrary to hypothesis; or some already actual part of it actualizes the potency, in which case that already actual part would *itself* be both pure act and, properly speaking, the true first mover. Now, having no potency to actualize, such a being could not possibly change or move. Thus we have reached a first mover that is not only unmoved, but unmovable.

MacDonald might object to this that the move from accounting for the activities or operations of things to accounting for their existence or being in effect involves an appeal to something other than motion, and thus to an

argument other than the argument from motion; and though (as MacDonald would acknowledge) this would not by itself show that there is anything wrong with the argument, it would leave untouched his claim that the First Way by itself is incomplete and "parasitic" for any effectiveness it has vis-à-vis proving God's existence on some other argument. But as commentators like Norman Kretzmann and D. Q. McInerny have noted, if the point of an argument from motion is to explain motion, and to explain motion requires explaining the existence of the things doing the moving and the way in which factors outside them contribute to their ability to move, then an explanation of the existence of moving things is quite naturally going to be a part of any argument from motion. More to the point, if motion is just the reduction of potency to act, then since the existence of a thing no less than its activity involves (in everything other than that which is pure act) the reduction of potency to act, any explanation of motion must account for the existence of things and not just their activities. Far from making an argument from motion "parasitic" on some other kind of argument, the move to the explanation of the existence of moving things is a necessary part of any such argument.

Notice that at no point in our exposition of the argument from motion have we had to appeal to any claims from Aristotelian physics, "outmoded" or otherwise. The argument proceeds entirely in terms of such metaphysical notions as the act/potency distinction, the principle of causality, and so forth. Still, it is sometimes suggested that Newton's principle of inertia undermines the proof from motion, for if (as that principle tells us) it is just a law of physics that a body in motion tends to stay in motion unless acted upon from outside, then (so it is claimed) Aquinas's view that whatever is moving must here and now be moved by something else is thereby shown to be false. But there are several problems with this objection. First and most obviously, Newton's principle concerns only "local motion" or movement from one place to another, while motion in the Aristotelian sense includes (as we have seen) not just local motion, but also changes in quality (like water becoming solid when it freezes), changes in quantity (as when a pool of water gets larger or smaller), and changes in substance (as when hydrogen and oxygen are combined to make water) (*In Phys* III.2.286). (There is a strict sense of "motion" within the Aristotelian tradition on which changes in substance are not counted as motions, but they are motions or changes in a loose sense; and as several commentators have noted, they do in any case count as reductions of potency to act of the sort the argument from motion seeks to account for.) At the very least, then, the defender of the First Way can say that whether or not local motion needs an explanation of the sort the argument provides, these other kinds of change do need such an explanation.

But in fact there is no good reason to exclude local motion from the range of that which needs explanation in terms of a first unmoved mover. After all, it is no good just to say "Well, it's simply a law of physics that things in motion tend to stay in motion unless acted upon from outside." For one thing, there is still the question of what puts something in motion in the first place, and in general of a thing's acquisition or loss of momentum, and explaining these events will require just the sort of explanation the First Way tells us other instances of change do. More fundamentally, we also still need to know *what it is* exactly for something to be a law of physics, and *why* such a law holds.

Regarding the first question, some defenders of the First Way have suggested that Newton's principle is nothing more than a mathematical model which is of utility in making predictions but which strictly speaking does not describe the objective nature of physical objects. One reason for adopting such an instrumentalist (as opposed to realist) interpretation of the principle of inertia is that to interpret the principle realistically would commit us (so it is argued) to the metaphysically absurd consequence that a finite cause can have an infinite effect. It is then sometimes further suggested that to explain local motion, especially of a projectile sort, we need therefore to postulate that the initial cause of a thing's movement (the arm which throws a spear, say) imparts to it a force, "impulse," or "impetus" which keeps it in motion, and thus passing from potency to act, as long as it does move, and where this impetus serves as an instrumental cause whose efficacy must ultimately be traced to the simultaneous activity of a first mover. Other defenders of the argument reject this "impetus" theory and would grant that Newton's principle does tell us something about the real nature of physical objects. But they would then insist that this simply leaves us with the question of what actualizes the potential existence of things having natures of the sort described by the principle of inertia, and that to answer this question we have (for reasons already seen) to appeal to something which is pure act. In short, Newton's principle can hardly undermine the First Way if the existence of a first unmovable mover is needed in order to explain why the principle holds in the first place.

But it may be that even these general points concede too much force to the objection, for things are much less conceptually clear cut here than it might at first appear. For example, if, as is standardly done, we think of Newtonian inertial motion as a "state" rather than a process, then we need to get clear on exactly how such "motion" could be motion in the Aristotelian sense (i.e. a genuine *change*), in which case it also needs to be made clear exactly how Newton's principle is supposed to conflict with the Aristotelian principle that what is in motion (that is, changing) requires a mover. Or if inertial motion *is*

change of some sort, then we need to get clear on the sense in which such motion can be said to be a "state." It should also be kept in mind that in the physical universe as it actually exists, no object undergoing local motion is ever unaffected by outside forces, given for example the constant gravitational attraction every body exerts on every other. Hence at every moment at which an object is moving through space, and not merely at its initial acquisition of momentum, its motion is being affected in a way that requires explanation in terms of something outside it. But in that case, even with respect to the explanation of local motion, the principle of inertia seems practically moot. The conceptual waters here are deep, and reflect difficulties for interpreting modern physics that arise *whatever* its relationship to Aristotelian metaphysics. The point is that those who assert a conflict between Aquinas and Newton simply have not made their case until they have worked out these crucial details. It will not do lazily to assert, without addressing these issues, that modern physics has somehow "explained" local motion in such a way that reference to a first mover is unnecessary.

Another objection sometimes raised against the First Way is that anything moving something else, including a first mover, would have to be undergoing motion itself, as for example the hand of our example moves even as it is moving the staff. Therefore (the objection continues) the very notion of an unmoved mover is incoherent. But this objection begs the question. The argument from motion claims to prove that no motion, including the motion of moved movers, would be possible at all unless there is a first mover which is pure act and thus unmovable. So, given that the premises of this argument are true and that the conclusion follows logically from them, it follows that the conclusion is true and therefore coherent. Accordingly, it won't do simply to insist that the conclusion must be false; one has to show specifically either that one of the premises is false or that the conclusion does not follow. Otherwise, one ought to admit that the argument shows precisely that an unmoved mover really is possible (since actual) after all.

Besides, it is hardly as if the notion of an unmoved mover were anything like as problematic as that of (say) an "immortal mortal." An "immortal mortal" would be something that both dies and does not die, which is self-contradictory. But an unmoved mover is something that makes other things move without itself undergoing motion, and there is no *obvious* self-contradiction in that. Furthermore, as G. H. Joyce argues, the reason that the movers of our experience are themselves moving even as they move other things is precisely because they are limited in the various ways entailed by being composites of act and potency. (For example, because an arm is actually at one point in space and only

potentially at another, its potential to be at some other point in space has to be actualized by something else if it is to get the staff to that other point in space.) But something which is pure act, devoid of all potency, would have no such limitations, and thus not need to be moved itself as it is moving other things. Moreover, it would (as we shall see later) be outside of time, and indeed that which creates time, so that to the extent that the objection in question implicitly assumes that the first mover goes from not acting at one moment in time to acting at another moment in time, the objection simply misconceives the nature of the first mover's activity (*In Phys* VIII.2.989). Finally, as Garrigou-Lagrange points out, given that (as we will see a little later on) our knowledge of the first mover is necessarily largely negative, it should not be surprising if it is harder for us to get our minds around it than it is for us to understand the more mundane movers of our experience.

We have devoted a good deal of space to the First Way, partly because of its intrinsic importance and partly because Aquinas himself put so much emphasis on it. (He famously regarded it as the "more manifest way" [*ST* I.2.3] and presented versions of the argument from motion again and again in his writings, as the citations given above indicate.) Moreover, many of the issues that arise in the discussion of the First Way, such as the impossibility of an infinite regress of causes ordered per se, also arise in discussion of the other ways. Hence our fairly detailed discussion of the First Way helps to set the stage for our treatment of the others. Most importantly, it has also (hopefully) shown that the objections commonly raised against the argument are hardly as conclusive as they are sometimes assumed to be, and that it is, accordingly, as worthy of consideration today as it was in Aquinas's day.

The Second Way

The proof from causality begins by noting that the senses reveal to us an order of efficient causes. But nothing can be the cause of itself, for if it were then "it would be prior to itself, which is impossible" (*ST* I.2.3). Now in a series of efficient causes, the first cause is the cause of the intermediate cause or causes, which are in turn the cause of the ultimate cause. So if there were no first cause, then there would be no intermediate or ultimate causes at all (and thus no causes of the sort we started out acknowledging that we know through the senses). But if the series of efficient causes regressed to infinity, then there would be no first cause. Hence the series cannot go on to infinity, and "therefore it is necessary to admit a first efficient cause, to which everyone gives the name of God" (*ST*

I.2.3).

Let us note first (and yet again) that Aquinas does *not* say, here or elsewhere, that "everything has a cause"; rather, he begins the argument by saying that there are efficient causes and that nothing can cause itself. The implication is that if something is caused, then it is something outside the thing being caused that is doing the causing; and as we have seen in chapter 2, Aquinas is committed in particular to the principle of causality, according to which *that which comes into being*, or more generally *that which is contingent*, must have a cause. Needless to say, this is not the same thing as to claim that *everything* without exception has a cause. So the argument is in no way vulnerable to the stock objection aimed at the stock caricature of cosmological arguments (i.e. "If everything has a cause, then what caused God?"). We have also already seen in chapter 2 how the principle of causality might be defended against the sorts of objections raised by Hume.

There is another sort of objection to the principle of causality, however, or at least to the application made of it by arguments like the Second Way. According to Immanuel Kant (1724–1804), the principle of causality applies only to the world of sensory experience and cannot take us beyond it to a transcendent first cause. As usually presented, however, this objection seems little more than a dogmatic refusal even to consider the possibility of a proof for a first cause; certainly no one who does not accept Kant's highly controversial conception of the nature and limits of human knowledge has any reason to take it seriously. More to the point, as Maurice Holloway argues, from the fact that our knowledge of the principle of causality derives from our experience of sensible things (i.e. things which can be sensed), it simply doesn't follow that it cannot be applied beyond the realm of sensory experience. For the principle applies to sensible things themselves not insofar as they are sensible, but rather insofar as they exist. In particular, a stone, tree, or human being stands in need of a cause not by virtue of being a sensible object, but rather by virtue of being something to whose essence an act of existing must be conjoined if it is to be real. Yet, as even most non-Thomists would acknowledge, the notion of existence is broader than the notion of the sensible; certainly there is no difficulty making sense of the idea of non-sensible existing things given Aquinas's doctrine of analogy, the transcendentals, and so on. Hence there is no reason to doubt that the principle of causality applies beyond the realm of sensible things. At any rate, simply to insist, on the basis of some non-Thomistic epistemology (e.g. a Kantian epistemology) that it cannot apply to non-sensible things, merely begs the question against Aquinas.

It should be obvious that the reason Aquinas rules out the possibility of an

infinite regress of causes in the Second Way is the same as the reason he rules out the possibility of an infinite regress of movers in the First Way. What he has in mind, here as there, is a causal series ordered per se, not a causal series ordered *per accidens*; and his point, accordingly, is not that the universe must have had a beginning in time, but rather that whether or not it has always existed, it must here and now be sustained by a first uncaused cause. Hence, the points made in exposition of the First Way vis-à-vis infinite causal regresses apply here as well.

Indeed, it might seem that the two arguments differ little except verbally, the one making reference to motion, the other to efficient causes, but in substance more or less saying the same thing (especially given that moving or changing something just is an instance of efficient causation). But of course, the fact that Aquinas bothers to present them as distinct arguments at all is a clue that there must be more than a verbal difference between them (otherwise he could have stopped with four ways rather than five). Some commentators have suggested that the substantive difference between them is that the First Way seeks to explain how the things of our experience are capable of being passive recipients of change, while the Second Way seeks to explain how they can be active agents of change. But this seems questionable given that the First Way speaks not only of how some things undergo change, but also of how other things can cause change (even if only by being instruments of the first cause), and that the Second Way speaks not only of how things can cause change, but also of how they are being caused. A more plausible and interesting account of the difference between the arguments is provided by Etienne Gilson, who suggests that whereas the First Way is concerned to explain why things undergo change, the Second Way is intended to explain why they exist at all, where (as in the First Way) the causal influence of the first cause is not something that occurred merely at some point in the past, but which exists here and now. That is to say, just as the First Way is meant to show that no motion or change would occur here and now unless there were a first unmoved mover operating here and now, the Second Way is meant to show that nothing would even exist here and now unless there were a first uncaused cause sustaining things in being here and now.

One way to understand this interpretation is in terms of an argument for God's existence that Aquinas presents in chapter 4 of *On Being and Essence*, and which is sometimes called "the existential proof" or "the existence argument." Recall that for Aquinas, in everything other than God, essence is distinct from existence. (This isn't to assume from the outset that God exists, an assumption which would of course make the argument that follows a circular one; the point is just that *if* there is a God – which at this stage of the argument is yet to be

determined – in him alone essence and existence would be identical.) So, how does a thing come into existence? That is to say, how is its essence conjoined with an act of existence so that it is made real? "It is impossible," Aquinas says, "that the act of existing itself be caused by the form or quiddity – and by 'caused' I mean as by an efficient cause – for then something would be the cause of itself and produce itself in existence, which is impossible" (*DEE* 4). In other words, a thing's essence, form, or quiddity cannot be what brings the thing into existence, for considered by itself an essence is merely potential, and thus cannot cause anything. For an essence to be able to cause something it would first have to be actualized by being conjoined to an act of existing, and that would entail that the thing itself (since it just is a composite of an essence with an act of existing) would already exist. Hence the essence of a thing could cause its existence only if the thing already existed, in which case the thing would in effect be bringing itself into existence, which is incoherent. "It is therefore necessary that everything whose act of existing is other than its nature have its act of existing from another" (*DEE* 4). But a series of things deriving their acts of existing from something else cannot go on to infinity. Hence "everything which exists through another is reduced to that which exists through itself, as to a first cause" and "there must be something which causes all things to exist, inasmuch as it is subsistent existence alone" (*DEE* 4). That is, there must be something whose essence and existence are identical, and this we call God.

Keep in mind that a thing's essence and act of existing are distinct not just before it exists, but always, even after they are conjoined so as to make the thing real. (To put a handle on to a brush so as to make a broom doesn't make the handle identical to the brush; neither does conjoining an essence and an act of existence make *them* identical.) Hence it is not enough for a thing to be real that its essence and act of existing be conjoined merely at some point in the past; the essence and act of existing must be *kept* together at every point at which the thing exists. Accordingly, a thing must be caused to exist not once for all, but *continuously*, here and now as well as at the time it first came into being; to use the traditional theological language, it must be *conserved* in existence from moment to moment. But if what conserves it in existence were something which itself was a composite of essence and existence, then that conserving cause would need to be conserved as well. Insofar as the existence of a thing in whom essence and existence are distinct might involve a series of causes, then, we are once again talking about a causal series ordered per se, and thus (as Aquinas says) a causal series which necessarily depends on a first member which is not conserved by anything, but simply exists. In the nature of the case, this could only be something whose essence and existence are not distinct (and thus in need

of being conjoined) but identical.

There are obvious affinities between this "existential proof" and the Second Way. Both are concerned with accounting for the existence of things here and now, both reject the notion that a thing could cause itself, and both argue that a series of efficient causes must terminate in a first cause. It is natural to suppose that Aquinas intended in the Second Way to summarize the argument from *On Being and Essence*. But William Lane Craig argues that reasoning of the sort represented by the "existential proof" cannot be smoothly assimilated to the Second Way, for two reasons. First, the Second Way is supposed to take as its starting point causal chains that are evident to the senses, but the conjoining of an essence to an act of existence is not something we can observe. Second, in Aquinas's view only God can possibly conjoin an essence and an act of existing, so that the question of a series of causes, whether infinite or finite, cannot even arise for an "existential proof" style of argument; for God's causal activity in this case would have to be direct rather than instrumental (*ST* I.45.5). Accordingly, for Craig the Second Way must be interpreted as concerned with more mundane respects in which one thing causes, here and now, the existence of another, such as the way in which your existence is here and now dependent in part on the temperature of the earth's atmosphere, which is in turn dependent on the earth's distance from the sun, and so on.

Yet Craig's suggestion is not altogether convincing. For one thing, it would be very odd, especially given the centrality of Aquinas's doctrine of essence and existence to his metaphysical system in general and to his conception of God in particular, if he left the argument of *On Being and Essence* off his list of the Five Ways in which he says God's existence can be proved; and the Second Way is the closest of the Five Ways to the argument of *On Being and Essence*. (The Third Way, as we shall see, does deal with the distinction between contingent and necessary beings, but for Aquinas this doesn't correspond to the distinction between beings in which essence and existence are distinct and beings in which they are identical.) Moreover, though Craig correctly notes that Aquinas believes that only God can conjoin essence and existence, this did not stop Aquinas himself from raising the issue of a series of causes in the course of giving the "existential proof" in *On Being and Essence*; in particular, he says that a first cause of the being of things is necessary, for "otherwise we would proceed to infinity in causes" (*DEE* 4), which, again, parallels the argument of the Second Way. Perhaps he did so purely "for the sake of argument"; that is to say, Aquinas may have meant to say only that even if there were a series of causes involved in the conjoining of the essence and existence of a thing (which he did not in fact think there is), such a series would still have to have a first member.

But if the notion of a series of causes was indeed being raised in the "existential proof" in this purely "for the sake of argument" manner, perhaps that is also the spirit in which Aquinas meant to raise it in the Second Way. Finally, while the Second Way does indeed explicitly begin with what is evident to the senses, so too does the First Way, in its appeal to the fact of motion; and yet the First Way also almost immediately makes a transition into metaphysics insofar as it analyzes motion in terms of act and potency. It does not seem unreasonable to suppose that in the Second Way too, though Aquinas begins with something evident to the senses, namely the fact that things are caused to exist, the rest of the argument is to be read in terms of the metaphysical analysis of a thing's existence as something that needs to be conjoined with its essence if the thing is to be real. This would certainly make it easier to understand why a first cause would have to be itself uncaused: it could not fail to be if its essence and existence are identical, and thus in no need of being conjoined by a yet further cause.

It seems plausible, then, to read the Second Way in light of the "existential proof" of *On Being and Essence*, an argument which Thomists have in any event always considered extremely important to a proper understanding of Aquinas's conception of God, his relationship to the world, and the grounds of our knowledge of his existence. And whether they are more or less the same argument or not, they both certainly bring into focus Aquinas's view that the way in which philosophy can reveal to us the existence of a divine creator is *not* by proving that God must have caused the world to exist at some point in the past (which of course would raise the question of whether he still exists), but rather by proving that God must be sustaining the world in being here and now, and at any moment in which we are considering the question of his existence. As Peter Geach puts it, for Aquinas the claim that God made the world "is more like 'the minstrel made music' than 'the blacksmith made a shoe'"; that is to say, creation is an ongoing activity rather than a once-and-for-all event. While the shoe might continue to exist even if the blacksmith dies, the music necessarily stops when the minstrel stops playing, and the world would necessarily go out of existence if God stopped creating it.

These considerations should make it clear why an objection to "first cause" arguments famously raised by Hume (and distinct from his criticism of the principle of causality) has no force against Aquinas's argument. If we have explained each member of a causal series by appealing to an earlier member, what need, Hume asks, do we have for a first cause? For even if we trace the series of causes back infinitely, we will never have a case where any individual thing is left unexplained. As we have seen, if causal series ordered *per accidens*

are in question, Aquinas would agree with Hume that no first cause is necessary. But it is causal series ordered per se that the Second Way, like the First Way, is concerned with, and here the need for a first cause follows from the fact that in such a series all causes other than the first cause are purely instrumental, having no causal power of their own at all. Extending the series back to infinity would not change this in the least; as noted in our discussion of the First Way, even if a causal series ordered per se *were* infinitely long, as long as each member of this infinitely long series was purely instrumental, and thus causally inert of itself, there would have to be a cause outside the series which imparted causal power to all of the series' members, a cause which would then be "first" not in the sense of coming at the head of the series, but rather in the sense of being that on which every member of the series depends for its causal power. The irrelevance of Hume's objection is even more obvious when we consider the role played in Aquinas's argument by the distinction between essence and existence. For even if each member in a causal series extending backwards in time was caused by some earlier member, with the series going back infinitely, as long as the existence of each member is distinct from its essence, it will have to be conserved in existence at each moment by a first cause in whom existence and essence are identical. What matters is what causes each member to exist *here and now*; causes existing at previous moments of time, even if they are infinite in number, are totally irrelevant.

Another, related objection to "first cause" arguments is that they allegedly commit the "fallacy of composition." If each brick in a certain wall weighs a pound, it doesn't follow that the wall as a whole weighs a pound; similarly (the objection continues) if each thing in the universe requires a cause, it doesn't follow that the universe as a whole must have a cause. But there are two problems with this objection, at least considered as a criticism of Aquinas. First, as is well known to logicians, part-to-whole reasoning of the sort under consideration is not in fact always fallacious. For example, if every brick in a wall built out of a child's Lego blocks is red, then it follows that the wall as a whole is red. Similarly, given that the distinction between essence and existence suffices to show that any particular material thing requires a cause, it is surely correct to say that the universe as a whole, which is comprised of these material things and which itself has an essence distinct from its existence, must also have a cause. Second, Aquinas's argument does not in fact require reasoning in this part-to-whole fashion in any case. To get the proof from causality going (especially if it is understood in light of the "existential proof") one need not consider the universe as a whole, but just any individual thing whose essence is distinct from its existence – a book, a car, a dog, a tree, whatever. For to explain

even that single thing will (if Aquinas's argument is otherwise unobjectionable) require appeal to something whose essence and existence are identical, and thus appeal to an uncaused first cause.

The Third Way

The proof from the contingency of the world starts from the fact that there are in the natural order things for which it is possible either to exist or not exist, as is evident from the fact that they are generated and corrupted, coming into being and passing away. But "that which is possible not to be at some time is not" (*ST* I.2.3); that is to say, if it is possible for something not to exist, then at some time it will not exist. "Therefore, if everything is possible not to be, then at one time there could have been nothing in existence" (*ST* I.2.3). Now if there ever were a time when nothing existed, then nothing would exist now, because "that which does not exist only begins to exist by something already existing" (*ST* I.2.3), so that if there was nothing in existence at some point in the past there would have been no way for anything new to be brought into existence. But since it would be absurd to hold that nothing exists now, it follows (given that the assumption that everything that exists is merely possible leads to this absurdity) that not everything that exists is merely possible, that is, capable of either existing or not existing; and therefore "there must exist something the existence of which is necessary" (*ST* I.2.3). Now "every necessary thing either has its necessity caused by another, or not" (*ST* I.2.3). But it is impossible to go on to infinity in a series of necessary things which get their necessity from another, for the reasons already discussed when considering series of efficient causes. Therefore there must be something "having of itself its own necessity, and not receiving it from another, but rather causing in others their necessity" (*ST* I.2.3), and this is what we call God.

In short, the Third Way holds that the world of contingent things could not exist at all unless there were a necessary being. It would be a serious mistake, however, to understand "contingent" and "necessary" here in the senses most familiar to contemporary philosophers, many of whom think (for example) of what is necessary as that which exists in every possible world and of what is contingent as that which exists only in some possible worlds, or who assume that the notion of a necessary being must be that of a being the denial of the existence of which would entail a self-contradiction (thus reading "necessary" as "logically necessary"). As we saw in chapter 2, Aquinas (like other Aristotelian essentialists) would not accept such modern accounts of necessity and

contingency.

One common objection to the Third Way which may reflect this misunderstanding is the suggestion that Aquinas commits an obvious fallacy when he claims that "that which is possible not to be at some time is not," for even if it is possible for something to go out of existence, it simply doesn't follow that it will actually do so. This objection would clearly be correct if by "possible not to be" Aquinas meant "non-existent in some possible world" or "the non-existence of which is logically possible," for it is obvious that neither the fact that there is a possible world in which something doesn't exist nor the fact that there is no self-contradiction involved in denying its existence entails anything about its longevity in the actual world. Similarly, it is sometimes claimed against cosmological arguments that only propositions can be necessary, and not things. This too might be a good objection to Aquinas if by "necessary" he meant "logically necessary." But again, Aquinas does not in fact mean "possible" or "necessary" in any of these modern senses, so these objections are irrelevant.

What Aquinas does mean is indicated by the reason he gives for saying that some things are possibly either existent or nonexistent, namely that we observe them to be generated and corrupted. Now as we saw in chapter 2, for Aquinas generation and corruption, coming into being and passing away, characterize the things of our experience because they are composites of form and matter. Their coming to be is just the acquisition by a certain parcel of matter of a certain form, and their passing away is just the loss by a certain parcel of matter of a certain form. Hence it is ultimately this composite, hylemorphic nature that makes it the case that they are "possible to be and not to be" (*ST* I.2.3); it has nothing to do with possible worlds, with there being no self-contradiction involved in denying their existence, or any other such thing. The "possibility" in question is not some abstract logical possibility but rather something "inherent," a tendency "to be corrupted" rooted "in the nature of those things … whose matter is subject to contrariety of forms" (*QDP* 5.3). In other words, given that the matter out of which the things of our experience is composed is always inherently capable of taking on forms different from the ones it happens currently to instantiate, these things have a kind of inherent metaphysical instability that guarantees that they will at some point fail to exist. They have no potency or potential for changeless, indefinite existence; hence they cannot exist indefinitely.

By "possible not to be," then, what Aquinas means is something like "having a tendency to stop existing," "inherently transitory," or "impermanent"; and by "necessary" he just means something that is not like this, something that is

everlasting, permanent, or non-transitory. Thus there is no fallacy in his inference from "such-and-such is possible not to be" to "such-and-such at some time is not," for this would follow given an Aristotelian understanding of the nature of material substances. Given enough time, such a substance would, if left to itself, *have* to go out of existence eventually. There is no sense to be made of the idea that it might be "possible" for it not to exist and yet that it never in fact goes out of existence no matter how much time passes and even if nothing acts to frustrate its tendency towards corruption, for in that case the claim that it has an inherent tendency towards corruption would be unintelligible. Something that *always* exists would by that very fact show that it is something whose nature does not include any inherent tendency towards corruption, and thus that it is necessary (*In DC* I.29).

However, this still leaves untouched an apparently more serious difficulty with the Third Way. Even if it is granted that Aquinas is justified in holding that whatever is "possible not to be" will at some time go out of existence, it is widely held that his further inference to the effect that if *everything* were "possible not to be" or contingent, then at one time nothing would have existed, is clearly fallacious. Specifically, it is claimed that he is guilty here of a "quantifier shift" fallacy, of inferring from "Everything has some time at which it does not exist" to "There is some time at which everything does not exist." This is called a "quantifier shift" fallacy because the quantifying expression "everything" shifts position from the first statement to the second. That it is a fallacy can be seen by comparing the argument above with parallel arguments that are clearly fallacious. If every student in the room owns a pencil, it does not follow that there is a certain pencil that every student in the room owns; if every human being has someone as a mother, it does not follow that there is someone who is the mother of every human being; and so forth. Similarly, even if every contingent thing goes out of existence at some time, it does not follow that there is some time when they all go out of existence together. An alternative possibility is that even though every contingent thing goes out of existence at some point, there is always at least one other contingent thing that continues to exist in the meantime, and this overlapping series of contingent things could continue on infinitely. (Certainly Aquinas could not rule such an infinite regress out, since it would involve a causal series ordered *per accidens* extending backward in time, and as we have seen, Aquinas concedes for the sake of argument that such a series might not have a first member.) In this case, though, Aquinas's conclusion to the effect that if everything were contingent then nothing would exist now would be blocked, and the Third Way would fail.

But common though this objection is, it is not in fact fatal to Aquinas's

argument, for he need not be interpreted as arguing in the fallacious manner described. As several commentators have suggested, what Aquinas really seems to be getting at is the idea that given an infinite stretch of time, and given also the Aristotelian conception of necessity and possibility described above, then if it is even possible for every contingent thing to go out of existence together (which even Aquinas's critic must concede), this possibility must actually come about. For (again, at least given an Aristotelian conception of possibility) it would be absurd to suggest both that it is *possible* for every contingent thing to go out of existence together, and yet that over even an *infinite* amount of time this will never in fact occur. "Possibility" here entails an inherent tendency, which must manifest itself given sufficient time, and an infinite amount of time is obviously more than sufficient. Hence if everything really were contingent, there would have been some time in the past at which nothing existed, in which case nothing would exist now, which is absurd, and so on, and Aquinas's argument would (up to this stage in the proof at least) be vindicated. (Note that it would not help the critic to suggest that the series of contingent things had a beginning in time after all rather than being infinite, for in that case Aquinas could simply say that given the principle of causality this beginning must then have had a cause and that this cause would have to be something non-contingent, i.e. necessary, which is of course what he has been trying to prove the existence of all along.)

At this point the critic of the Third Way might think to challenge the premise that "that which does not exist only begins to exist by something already existing," so as to undermine the claim that if there was ever a time when nothing existed, then nothing would exist now. But this premise is just a variation on the principle of causality, and we have already seen how that principle might be defended. A more promising strategy for the critic might seem to be to suggest (as J. L. Mackie does) that even if individual contingent things all go out of existence, there might still be some underlying stuff out of which they are made (a "permanent stock of matter," in Mackie's words) which persists throughout every generation and corruption. Now if this were so, then what would follow, given the Aristotelian conception of necessity we've been describing, is that this stock of material stuff would itself count as a necessary being. But (so the suggestion continues) the critic could happily accept this (as Mackie does) given that such a "necessary being" would, in view of its material nature, clearly not be divine.

The trouble with this reply, though, is that it falsely purports to be asserting something that Aquinas would deny. In fact, surprising as it might seem, Aquinas would be quite happy, at least for the sake of argument, to concede that

the material world as a whole might be a kind of necessary being, in the relevant sense of being everlasting or non-transitory. After all, as we have repeated many times, Aquinas does not think that proving the existence of God requires showing that the material world had a beginning. Moreover, as we noted in our discussion of hylemorphism in chapter 2, Aquinas himself insists that while individual material things are generated and corrupted, matter and form themselves are (apart from special divine creation, to which he would not appeal for the purposes of the argument at hand lest he argue in a circle) not susceptible of generation and corruption. Far from regarding the notion of the material world as necessary as a blow to the project of the Third Way, Aquinas would in fact regard it as a vindication of his claim that there must be a necessary being. Indeed, he recognizes the existence of other non-divine necessary beings as well, such as angels and even heavenly bodies (which, given the astronomical knowledge then available, the medievals mistakenly regarded as not undergoing corruption).

That this should not be surprising, and in particular that it should not be regarded as damaging to the aim of proving the existence of *God* specifically, should be evident when we remember that proving the existence of a necessary being is only one component of the overall argumentative strategy of the Third Way. For recall that at this stage of the argument Aquinas immediately goes on to say that "every necessary thing either has its necessity caused by another, or not" and then argues that a series of necessary beings cannot go on to infinity. This might seem very odd to those contemporary philosophers who think of necessity in terms of possible worlds or who regard all necessity as logical necessity. "How could a necessary being get its necessity from another?" such a philosopher might ask. "It either exists in all possible worlds or it does not, or the assertion of its non-existence either involves a self-contradiction or it does not. End of story. Certainly there can be no question of anything *causing* it to exist in all possible worlds or *causing* it to be logically necessary!" But when we keep in mind that Aquinas does not mean "necessary" in the sense in which such contemporary philosophers understand it, but rather in the sense of "everlasting" or "permanent," we can see that it makes perfect sense to consider whether a thing's necessity is derived or not. In particular, we can see that it is not enough to show that the material universe as a whole (or an angel, a heavenly body, or whatever) is a necessary being in the relevant sense. One also needs to know whether it is the sort of thing that could possibly have its necessity in itself, or whether instead it must derive its necessity from something else, from something which *keeps* it in existence everlastingly.

It is immediately obvious, however, that matter qua matter cannot possibly

have its necessity of itself, at least on an Aristotelian conception. For matter considered apart from anything else, and in particular apart from form, is just "prime matter" or pure potentiality; and pure potentiality, since by definition it has no actuality, has no reality either, necessary or otherwise. Matter exists only insofar as it is combined with substantial form to comprise a substance. Nor would it help the critic of the Third Way to suggest that it is matter and form together that constitute a necessary being having of itself its own necessity. For one thing, and as we have already noted, individual material things are constantly going out of existence and thus losing their forms, and it is in their nature to do so. Hence it cannot be any particular material substance, but only prime matter, which can be said to be everlasting (and prime matter, for the reasons just given, cannot have its everlastingness of itself). Second, even if there could be some composite of form and matter which exists everlastingly, since in purely material substances form depends on matter just as matter depends on form, we would have (as Martin has pointed out) an explanatory vicious circle unless we appealed to something outside the form/matter composite on which it depends for its existence. Third, since (given Aquinas's doctrine of essence and existence) the existence of any material thing is distinct from its essence, we would need in any case to appeal to something outside it in order to explain how its essence and existence come together so as to make it real. (Note that this particular point would apply to material things even if, contrary to Aristotle and Aquinas, we did not regard them as composites of form and matter.) There is no way, then, plausibly to hold that matter might have its necessity of itself. Even a "necessarily existing" or everlasting material world would have to depend on something outside it for its existence. And this something could not itself be a composite either of form and matter or essence and existence, on pain of infinite regress.

The essence/existence distinction also implies that other sorts of non-divine necessary beings, such as angels (which on Aquinas's view are composites of a pure form together with an act of existing), would have to derive their necessity from something else. The only thing that could stop an explanatory regress of necessary beings would therefore be something whose essence and existence are identical, and who is a necessarily existing being precisely because it just *is* subsistent being or existence. Here we need only refer back to the "existential proof" considered when discussing the Second Way in order to fill in the details; and the upshot is that the Second Way and Third Way appear to converge on exactly the same being, albeit they do so from very different starting points (and thus remain distinct arguments).

One serious weakness of Mackie's otherwise reasonably fair-minded

discussion of the Third Way (in the context of what is possibly the best book in philosophy of religion written from an atheistic point of view, *The Miracle of Theism*) is that he never considers the relevance of Aquinas's hylemorphic conception of matter or distinction between essence and existence. Hence he mistakenly concludes that the only way Aquinas could show that the material world itself is not the ultimate necessary being is to transform the Third Way into something like Leibniz's cosmological argument and define a necessary being as one which exists in all possible worlds, or one the non-existence of which would involve a logical self-contradiction, or one whose essence "involves" or "includes" existence. But (to repeat yet again) Aquinas does not mean "necessary" in either the "possible worlds" sense or the logical sense. Nor does he think that a necessary being having its necessity of itself is one whose essence "includes" existence, as if its existence were simply one attribute it had alongside others. Rather, he thinks of it as something which just *is* subsistent existence, Being Itself rather than "a being" among other beings, and (as we shall see later on in this chapter) something absolutely simple or non-composite in which no distinctions can be drawn between its various attributes.

In general, it is remarkable how many critics of the Five Ways almost completely ignore Aquinas's own metaphysical views, and instead read into the arguments all sorts of assumptions that Aquinas himself never made and often would have rejected. It is no wonder, then, that (as we have seen) the objections these critics raise are often wide of the mark. If this is so in the case of the first three Ways, it is perhaps even more so of the last two, to which we turn next.

The Fourth Way

The proof from the grades of perfection begins by noting that "among beings there are some more and some less good, true, noble, and the like" (*ST* I.2.3). But things are said to be "more" or "less" a certain way to the extent that they "resemble" some maximum, "as a thing is said to be hotter according as it more nearly resembles that which is hottest" (*ST* I.2.3). But in that case, it follows that "there is something which is truest, something best, something noblest, and, consequently, something which is uttermost being; for those things that are greatest in truth are greatest in being" (*ST* I.2.3). Now the maximum within any genus is the cause of everything in that genus, "as fire, which is the maximum of heat, is the cause of all hot things" (*ST* I.2.3). So there must be something which is the cause of the "being, goodness, and every other perfection" of all beings, and this is what we call God (*ST* I.2.3).

Of all of the Five Ways, the fourth is generally regarded as the most difficult for modern readers to accept, or even to understand. Even Christopher Martin, whose reading of the other four Ways is very sympathetic, does not claim to understand it, and judges it "strange" or even "bizarre." It seems to me, though, that the mysteriousness of the Fourth Way has been greatly exaggerated, and that while it may well be more out of sync with contemporary philosophers' metaphysical predilections than Aquinas's other proofs, it is perfectly comprehensible and even defensible when properly understood in light of his general metaphysical commitments.

The argument is often said to be among the most Platonic elements of Aquinas's thought, and while it does not in fact presuppose the truth of Platonism, reading it in light of Plato's Theory of Forms does provide at least a useful first approximation to what Aquinas is getting at. For Plato, the ordinary objects of our experience can only be understood in terms of their "resemblance" to or "participation" in ideal archetypes of which they are but imperfect copies. To take a simple example, consider several triangles, some drawn in chalk on a board, some drawn in sand, some drawn on paper in pencil and others in various colors of ink. Now the essence or nature of a triangle is to be a closed plane figure with three straight sides, and it is by reference to this essence that we judge the particular triangles of our example to be triangles in the first place. But notice that each of these particular triangles is going to have certain features that have nothing to do with this essence; for example, some of them will be red and some green, some large and some small, some made out of chalk dust and some out of sand, and so forth, even though there is nothing about triangularity per se that entails any of these features. Notice too that they are all also going to lack, to some extent, some of the features that are part of the essence of a triangle. For instance, some of them will be drawn with partially broken lines or corners that are not perfectly closed, and none will be drawn with lines that are perfectly straight. Moreover, there are certain geometrical truths about triangles, such as that their angles add up to 180 degrees, that are necessary truths in the sense that they would remain true even if every individual material triangle went out of existence. Yet if any material triangle is going to have features that are not part of triangularity and will lack features that are part of it, and if there are truths about triangles that would remain true regardless of whether any material triangles actually exist, then triangularity per se – the essence or archetype by reference to which we judge something to be a more perfect or less perfect instance of a triangle, and indeed to count as a triangle at all – cannot be something material.

Neither, in Plato's view, can it be something mental. For the necessary truths

that we know about triangles (such as that their angles add up to 180 degrees, the Pythagorean theorem, etc.) are *objective* truths, something we *discover* rather than invent. We could not change them if we wanted to, and this shows that they do not depend for their existence on our minds. If triangularity as such is neither material nor mental, however, then it has a unique kind of existence of its own, that of an abstract object existing in a "third realm." And what is true of triangles is also true in Plato's view of more or less everything else: of circles, squares, and other geometrical figures; of human beings, dogs, cats, and other living things; of tables, chairs, rocks, trees, and other physical objects; of justice, goodness, beauty, piety, and the like; and so on. When we grasp the essence of any of these things, we grasp something that is *universal* rather than particular (since it is that in virtue of which various individual things count as instances of the same one type), *perfect* rather than imperfect (since it is the pattern or archetype by reference to which we judge something to be more or less perfect), and *eternal* or unchanging (since the truths we know about these essences are necessary truths). For these reasons we also thereby know something that is *more real* than individual particular things, since the latter only have their reality to the extent that they resemble or participate in the former. In short, what we know is what Plato calls a Form.

Now it is easy to see why the Fourth Way would seem to many readers to be Platonic in spirit. Aquinas argues that "'more' and 'less' are predicated of different things, according as they resemble in their different ways something which is the maximum" (*ST* I.2.3), and this "principle of exemplarity" (as Henri Renard has labeled it), with its talk of things "resembling" some maximum more or less perfectly, is certainly reminiscent of Plato. As is well known to scholars of his thought, Aquinas also frequently makes use of the concept of "participation," including in cases where he restates the principle of exemplarity (e.g. *QDP* 3.5, *ST* I.44.1, and *ST* I.79.4), and this concept is clearly Platonic in origin. And unlike the rest of the Five Ways, the Fourth Way seems (at least in its first stage) to be concerned with explaining the world in terms of formal rather than efficient causality (a distinction explained in chapter 2), another apparent departure from Aquinas's usual Aristotelian orientation towards a more Platonic one. The thrust of the argument might therefore seem to be that we can only make sense of the more or less good, true, and noble things of our experience by reference to something like a divine Platonic archetype of goodness, truth, and nobility.

A Platonic reading of the Fourth Way also has the advantage of forestalling an objection commonly heard in these more relativistic times, to the effect that standards of goodness, truth, nobility, and so on are all subjective. For if

Platonism is true, then such relativism and subjectivism are no more plausible in the case of goodness and the like than in the case of mathematics. Furthermore, the apparent emphasis on formal rather than efficient causality might seem to explain why Aquinas thinks that the maximum in any genus is the "cause" of everything in that genus. This claim is odd and implausible (so it is said) if Aquinas has efficient causality in mind, but understandable and defensible if he is speaking instead of formal causality.

On the other hand, there is one glaring problem with a Platonic interpretation of the Fourth Way, which is that Aquinas was not a Platonist, but rather an Aristotelian or moderate realist. That is to say, he did not believe in a realm of Forms or abstract objects existing altogether outside the world of concrete objects; as we saw in chapter 2, he took the forms of things to exist instead in the things themselves, and to exist in a universal and abstract way only in the intellect. And as it happens, this moderate realism is (as we shall see in chapter 5) perfectly sufficient to allay any concerns about the purported subjectivity or relativity of standards of goodness and the like; no appeal to Platonism is necessary. Moreover, even if an appeal to formal rather than efficient causality would solve one problem, it would raise another. For being abstract rather than concrete objects, Platonic Forms are causally inert (where efficient causality is concerned); hence if the Fourth Way were really suggesting that we think of God as a kind of Platonic Form, it would be hard to see how the most true, good, and noble being of the Fourth Way could be identical to the First Mover and First Cause of the first two ways.

Then there are certain other objections sometimes raised against the Fourth Way which would seem to be if anything only exacerbated by a Platonic reading. For example, why assume that the most true being, the most good being, and the most noble being are the *same* being? (After all, in Plato's thought each thing's Form is distinct from other Forms.) And given the reasoning of the Fourth Way, wouldn't we have to say that God is not only the most good, true, and noble thing, but also (to quote Dawkins again) the "perfect maximum of conceivable smelliness," and indeed that he possesses to the maximum degree any attribute we can think of? (After all, on Plato's theory, everything has a Form, including not only goodness, truth, and the like, but less elevated and abstract things too, like sweetness, filthiness, illness, and the like.) But this would be absurd, and certainly incompatible with Aquinas's conception of God.

In fact these objections, like others we've examined, rest on egregious misunderstandings of Aquinas's basic metaphysical commitments; and while there are indeed Platonic aspects to the Fourth Way, they are all greatly transformed by Aquinas in light of some of the concepts we surveyed in chapter

2, in a direction more consistent with his general Aristotelianism.

Let us note first that Aquinas is not in fact trying to argue in the Fourth Way that *everything* that we observe to exist in degrees (including heat, smelliness, sweetness, etc.) must be traceable to some single maximum standard of perfection. Here (as elsewhere in the Five Ways) his archaic scientific examples have led modern readers to misread him. Given the (false, we now know) medieval theory that fire is the source of all heat, he naturally appeals to fire and heat merely to *illustrate* the general principle that things that come in degrees point to a maximum. But heat itself is not among the things he is trying here to explain. (This should be obvious when you think about it, since Aquinas would clearly not regard heat or fire as divine attributes!) Rather, he intends to use the principle in question to explain *truth*, *goodness*, *nobility*, *being* and the like specifically. As the reader may have noticed, this list is very similar to the list of "transcendentals" we discussed in chapter 2, which are (unlike heat, smelliness, etc.) above every genus and common to every being, unrestricted to any particular category or individual. And as commentators on the Fourth Way generally recognize, Aquinas is mainly concerned in this argument to show that to the extent that these *transcendental* features of the world come in degrees, they must be traceable to a maximum. (It is true that "nobility" was not on the list of transcendentals we examined in chapter 2, but as Wippel points out, Aquinas's linking of *nobilitas* with being and perfection in *SCG* I.28 indicates that he does not think of it as a transcendental distinct from the others.) Since Aquinas is not in this argument concerned with heat, cold, sweetness, sourness, fragrance, smelliness, and other mundane features of reality, Dawkins' objection simply misses the point. Moreover, it should now be clear why Aquinas takes the most true, most good, and most noble being to be one and the same being; for as we saw in chapter 2, Aquinas argues that the transcendentals are "convertible" with one another. That is to say, they are one and the same thing considered under different descriptions. This is also why he draws a related inference that might otherwise seem ungrounded to many modern readers, to the effect that that which is most true, good, and noble is "consequently, something which is uttermost being; for those things that are greatest in truth are greatest in being." For this follows automatically from the doctrine of the transcendentals.

We also saw in chapter 2 that Aquinas takes different aspects of reality all to have being in an analogical rather than univocal sense. Accidents and substances both have being, but a substance has independent existence in a way accidents do not; material things and angels both have being, but angels (since they lack matter and are composed of pure form together with an act of existence) are metaphysically simpler than material things and lack the tendency towards

corruption that material things possess; created things and God both have being, but in created things essence and existence are distinct and in God they are not. Again, the way in which each has being is analogous to the way the others do, but not identical. In particular, it should be evident that substances have a higher degree of being than accidents do, angels a higher degree of being than material things do, and God a higher degree of being than any created thing; for substances lack the dependence on (other) substances that accidents have for their being, angels lack the dependence on matter that material things have for their being, and God depends on nothing at all for his existence but is rather that on which everything else depends. We see here a hierarchy in the order of being that dovetails with the hierarchy from prime matter through purely material things, human beings, and angels, up to God as Pure Act that we also had reason to discuss in chapter 2.

Given the convertibility of the transcendentals, it should not be surprising that, just as being does, goodness, truth, and the like come in degrees and are predicated of things analogically. For instance, the goodness or perfection of a triangle drawn carefully on paper with a pen and ruler is greater than that of a triangle drawn hastily in crayon on the cracked plastic seat of a moving bus, for it more perfectly instantiates the form or pattern definitive of triangularity. The goodness or perfection of someone who always tells the truth is greater than that of a habitual liar, for the former sort of person more perfectly fulfills the natural end or final cause of our intellectual and communicative faculties, which is to grasp and convey truth. A triangle and a person are both "good" in an analogical rather than a univocal sense, however, since there is a moral component to human goodness that is absent in the case of triangles and other non-rational entities. Moreover, human beings and triangles, along with other inanimate material things, plants, and non-human animals, manifest different degrees of goodness. Inanimate material things have certain perfections, such as (again) the straightness with which the sides of a triangle might be drawn. Plants, the simplest living things, have these sorts of perfections too given that they are material things, but in addition they have perfections that inanimate things do not have, namely the capacity to take in nutrients, grow, and reproduce themselves. Animals incorporate both the perfections of inanimate material things and plants, but in addition have the capacity for locomotion and sensation, which plants do not have. Human beings possess the perfections that inanimate material things, plants, and other animals have, but in addition have the capacity for intellect and will. Each of these levels of material being represents a higher level of goodness or perfection than the preceding one because it incorporates the perfections of the lower levels while adding perfections of its own. When we

get to the purely immaterial levels of the hierarchy of being, we have entities which, though they lack the perfections of material things "formally," they nevertheless possess them "eminently" insofar as (unlike purely material things on Aquinas's view) they can grasp them intellectually (and grasp them intellectually in a way that is superior to our way of grasping them, since though the human intellect is immaterial, it is limited because of its dependence on sense organs).

We will have reason to explore some of these matters in more detail in our next chapter, but the point for now is to indicate the way in which Aquinas takes the degrees of goodness, being, and the like to point to a single maximum. The idea is that if we start by considering the natures of each of the lower levels of reality and then proceed to follow them upward, we find ourselves inexorably led to a highest level. In particular, degrees of goodness, truth, nobility, and so forth each point beyond themselves to a highest degree of each; since these are all convertible with one another, it is the *same* one maximum to which they all point; and since they are all in turn convertible with being, this single maximum is also that which is most fully real. What Aquinas is up to in the Fourth Way can therefore be understood when we read the argument in light of his doctrines of the transcendentals, analogy, and the hierarchy of being.

In what sense is this highest level of reality the "cause" of the lower levels? And in what sense do the latter "participate" in the former if it is not in a Platonic sense? The answers to these questions are related. Something "participates" in a certain perfection when it has that perfection only in a partial or limited way (*In DH* 2); and for Aquinas, "whatever is found in anything by participation, must be caused in it by that to which it belongs essentially" (*ST* I.44.1). Unlike Plato, whose emphasis is exclusively on what later thinkers would call formal causality, Aquinas takes there to be an essential link between participating in something and being *efficiently* caused by it. How so? Consider first the specific case of existence or being, where we have already seen that for Aquinas, "from the fact that a thing has being by participation, it follows that it is caused" (*ST* I.44.1). The reason for this was that if a thing's essence and existence are distinct (so that it only "participates" in being or existence rather than being identical with pure being or existence), only something outside the thing could give it existence or being; for to say that its existence derives from its essence (which is the only other alternative) would entail the absurdity that it causes itself. (Keep in mind that deriving or flowing from an essence is not the same as being identical with an essence; for example, the essence of a human being is to be a rational animal, and having the capacity for language flows or derives from this essence, but having the capacity for language is nevertheless

not *identical* with being a rational animal.) We have also seen that, for Aquinas, the cause in question must ultimately be something in which its essence and existence are identical, and which accordingly just *is* being itself, or (we might now say) *unparticipated* being.

That, of course, is the heart of the "existential proof" and thus (I have suggested) the Second Way, which we have already examined. But given that being is convertible with goodness, truth, and the like, we would expect that what is true of things which have being or existence only by participation will also be true of things having goodness, truth, and so on only by participation, thus opening the way to a distinct argument for God's existence (namely the Fourth Way). And that is precisely what Aquinas thinks. In particular, he holds that in general (and not just with respect to being or existence) things that have some perfection only to various limited degrees must not have that perfection as part of their essence, "for if each one were of itself competent to have it, there would be no reason why one should have it more than another" (*QDP* 3.5). That is to say, if it were part of a thing's very *essence* to have the perfection, then there would be no reason for it not to possess it in an *unlimited* way. (Hence any human being is fully human, which follows from humanity being part of his or her essence, but does not have being to the fullest extent – which would be possible only for something whose essence just is being – or goodness to the fullest extent – which would be possible only for something in some sense having within it *every* perfection – and so forth.) So, for a limited thing to have some perfection, it must derive it from something outside it. And as Wippel notes, we would be led into a vicious infinite regress of the sort Aquinas has already criticized unless this something either is or is traceable to a cause which has the perfection to an unlimited degree.

But if the ultimate cause is unlimited in goodness, truth, nobility, or whatever other transcendental we are starting with, then (as we have already said) given the convertibility of the transcendentals it will also have to be unlimited in being and therefore just *be* pure being or existence itself. We are led therefore to the existence of the same being arrived at at the end of each of the first three ways – pure act, a being whose essence just is existence and which is the efficient cause of the being or actuality of everything other than itself – via yet another route, a consideration of the degrees of perfection found in the things of our experience.

The Fifth Way

The proof from finality starts with the observation that "things which lack

intelligence, such as natural bodies, act for an end, and this is evident from their acting always, or nearly always, in the same way, so as to obtain the best result" (*ST* I.2.3). From this it is plain that they act "not fortuitously, but designedly" (*ST* I.2.3). But whatever lacks intelligence can only act for an end if it is directed by something which has intelligence, "as the arrow is shot to its mark by the archer" (*ST* I.2.3). "Therefore some intelligent being exists by whom all natural things are directed to their end; and this being we call God" (*ST* I.2.3).

Aquinas's first three Ways are all variations on what is known as the "cosmological argument" for the existence of God (from the Greek *kosmos* meaning "order"). The Fourth Way is sometimes called the "henological argument" (from the Greek *hen* or "one"). The Fifth Way, in turn, is commonly taken to be a version of the "teleological argument" (from the Greek *telos* meaning "end" or "goal"). Etymologically speaking, this is an apt name for the proof, but it is also potentially misleading given that when most contemporary philosophers hear the expression "teleological argument" they naturally think of the famous "design argument," associated historically with William Paley (1743–1805), and defended today by "Intelligent Design" theorists critical of Darwin's theory of evolution by natural selection. Indeed, many writers (such as Richard Dawkins) assume that the Fifth Way is just a variation on the "design argument." But in fact Aquinas's argument is radically different from Paley's, and the standard objections directed against the latter have no force against the former.

Paley's argument was roughly this. Like some human artifacts, the universe is extremely complex and orderly; and while it is theoretically possible that this complexity and order was the result of impersonal natural processes, it is far more likely that it is the work of an intelligent designer. Paley's favorite examples of complexity and order are living things and their various organs. His successors in the "Intelligent Design" movement, though they attempt to formulate their position with greater mathematical rigor than Paley did, have followed him in this emphasis, focusing as they do on the purported "irreducible complexity" of various biological structures. Critics of the design argument respond that this is "God of the gaps" reasoning of the sort that is constantly vulnerable to being overthrown by the latest scientific research, which may well reveal (as it has in the past) that what seems at first glance to be irreducibly complex can be accounted for in terms of more simple, and impersonal, natural processes.

Whatever side one takes in this debate, it is irrelevant to the evaluation of Aquinas's Fifth Way, which differs from the design argument of Paley and the "Intelligent Design" movement in several crucial respects. Paley's argument

would justify, at most, belief in a deistic god who gave order to the world at some point in the past but who need not be appealed to in order to explain its current operation, which can be accounted for entirely in terms of impersonal laws of nature. "Intelligent Design" theorists even acknowledge that their arguments do not necessarily imply a deity at all, but merely a superhuman intelligence of some sort or other. Aquinas, by contrast, takes the Fifth Way to entail the existence of nothing less than the God of classical theism, who sustains the order of the world *here and now* and at any moment at which it exists. Moreover, while Paley and his contemporary successors claim only that the existence of a designer is *probable*, Aquinas takes the Fifth Way *conclusively to establish* the truth of its conclusion. Related to this, whereas the design argument is typically presented as a kind of quasi-scientific empirical hypothesis, Aquinas's argument is intended as a metaphysical demonstration. His claim is not that the existence of God is *one* possible explanation among others (albeit the best) of the order that exists in the universe (which is how "God of the gaps" arguments proceed) but rather that it can be seen on analysis to be the *only* possible explanation *even in principle*. While Paley and his successors focus on complex biological structures, Aquinas is not especially interested either in biology or complexity per se; even extremely simple inorganic phenomena suffice in his view to show that a Supreme Intelligence exists. Hence, while Darwin's theory of evolution by natural selection is notoriously problematic for the design argument, it is totally irrelevant to the Fifth Way. (That is not to say that Aquinas would agree that *every* aspect of the biological realm can be explained in the materialistic terms favored by Darwinians; as we will see in the next chapter, he would definitely not agree with this. The point is just that the debate over evolution is not relevant to the *Fifth Way* specifically.) And all of these dissimilarities derive ultimately from one key difference between the design argument and the Fifth Way, which is that whereas the former takes for granted a "mechanical" conception of the natural world of the sort early modern philosophers and scientists sought to put in place of Aristotelian teleology, Aquinas's argument crucially presupposes that final causes are as real and objective a feature of the natural world as gravity or electromagnetism.

We saw in chapter 2 how the reality of final causes might be defended today. We also saw that the sense in which teleology pervades the natural world on Aquinas's view is that efficient causes would not be intelligible without final causes. This is what he means when he says in the Fifth Way that "things which lack intelligence, such as natural bodies, act for an end, and this is evident from their acting always, or nearly always, in the same way, so as to obtain the best result." He is not especially interested here in the fact that hearts typically pump

blood, that eyes enable us to see, and other such biological facts (though these would naturally be included as instances of the more general phenomenon he is interested in). It is the existence of *any causal regularities at all* that he takes to require explanation (where the emphasis here, unlike in the Second Way, is on the "regularities" part of this phrase rather than the "causal" part). For Aquinas, the fact that A regularly brings about B, as B's *efficient* cause, entails that bringing about B is in turn the *final* cause of A. For if we did not suppose that A inherently "points to" or is "directed towards" the generation of B as its natural end, then we would have no way to account for the fact that A typically does generate *B* specifically, rather than C, or D, or E, or indeed rather than no effect at all. Of course, some interfering factor might prevent A from bringing about its typical effect, or from bringing it about fully or perfectly; this is why Aquinas speaks of a cause bringing about the "best" or perfect result at least "*nearly* always." But these unusual cases can only be understood against the background of the typical case, and in particular in light of the fact that a cause inherently *points to* the best or most perfect realization of its effect, even if it might sometimes be prevented by circumstances from bringing it about.

When Aquinas says that natural bodies do not bring about their effects "fortuitously," then, he is not arguing (as advocates of the design argument might) that it is improbable that complex structures could arise by chance, which would invite the response that natural selection shows how such structures might nevertheless arise by non-fortuitous but impersonal processes. For, to repeat, he is not interested here in complexity per se in the first place; as Garrigou-Lagrange points out, even a simple physical phenomenon like the attraction between two particles would suffice for his purposes. What he is saying is rather that it is impossible that *every* apparent causal regularity can be attributed to chance, for chance itself presupposes causal regularity. To take a stock example, for Aquinas, a paradigmatically fortuitous event would be a farmer's discovery of treasure under the ground he is plowing. Such a discovery was not in any sense intended – neither by the farmer nor by the person who buried the treasure, and not by nature either insofar as there is no causal law entailing that treasure will tend to be uncovered when one plows the ground. All the same, the farmer did intend to plow the ground, someone did intend to bury the treasure, and there are all sorts of causal laws operative when the farmer happens fortuitously to uncover treasure. Hence chance presupposes a background of causal factors which themselves neither have anything to do with chance nor can plausibly be accounted for without reference to final causality, so that it would be incoherent to suppose that an appeal to chance might somehow eliminate the need to appeal to final causality.

Given what Aquinas says about chance (and as Garrigou-Lagrange has also pointed out), it is a mistake to think that the "principle of finality" on which Aquinas's argument rests says that "*everything* has a final cause" (just as, as we have seen, it is a mistake to assume that the principle of causality says that "everything has an efficient cause"). For not "everything" does have a final cause, given the existence of chance events. What Aquinas actually says, as we have seen, is that every *agent* has a final cause; that is to say, that everything that serves as an efficient cause "points to" or is "directed at" some specific effect or range of effects as its natural end. This is why it is silly to ask (as is sometimes done) "What is the purpose of a mountain range?" or "What is the purpose of an asteroid?" as if such questions must be an embarrassment to any Aristotelian. Aquinas would be happy to allow that such things might turn out to serve no "purpose," in the sense of being accidental byproducts of convergent natural processes (plate tectonics or volcanism in the former case, say, and collisions between larger celestial bodies in the latter). He would insist, however, and quite plausibly, that such natural processes embody patterns of efficient causation that are themselves intelligible only in terms of final causation. And precisely for that reason, to the extent that biological processes like evolution manifest causal regularities, they if anything only *support* the Fifth Way rather than undermine it. For as with mountains, asteroids, and the like, even if it should turn out that animal species are the accidental byproducts of various convergent impersonal causal processes, the existence of those evolutionary processes themselves would require explanation in terms of final causes.

As these considerations together with those examined in chapter 2 indicate, then, the reality of final causes, in both the inorganic and organic realms, is as defensible today as it ever was. But if there really are final causes, then the first part of Aquinas's Fifth Way is vindicated. But what of the second stage of the argument, which claims that unintelligent natural processes can only act for an end if directed by some intelligence? The first thing to say is that this is not, as it might at first seem, a mere variation on the sort of reasoning represented by Paley's design argument. Paley, taking for granted as he does a modern mechanistic view of nature, denies that purpose or teleology is *immanent* or *inherent* to the natural order. That is why his argument is a merely probabilistic one. The design argument allows that there *might* in fact be no purpose at all in the natural world, but only the misleading appearance of purpose; its claim is simply that, at least where complex mechanistic processes are concerned, this supposition is unlikely. And even if there is purpose, it is imposed from outside, in just the way a human watchmaker imposes a certain order on metal parts that have no *inherent* tendency to function as a timepiece. The natural world remains

as devoid of immanent teleology after the designer's action as before. Moreover, as with a watch, once Paley's designer has done his "watchmaking," there is no need for him to remain on the scene, for once built the mechanism can function without him.

There is nothing like this kind of reasoning going on in Aquinas's Fifth Way. Like Aristotle, Aquinas takes the teleology or final causality that exists in nature to be immanent to it, to such an extent that one could for practical purposes (and as Aristotle himself did) ignore the idea of a designer altogether when searching out the final causes of things in the course of doing physical and biological science. (Note how different this is from the approach of contemporary "Intelligent Design" theorists.) A struck match generates fire and heat rather than frost and cold; an acorn always grows into an oak rather than a rosebush or a dog; the moon goes around the earth in a smooth elliptical orbit rather than zigzagging erratically; the heart pumps blood continuously and doesn't stop and start several times a day; condensation results in precipitation which results in collection which results in evaporation which in turn results in condensation; and so forth. In each of these cases and in countless others we have regularities that point to ends or goals, usually totally unconscious, which are just built into nature and can be known through observation to be there whether or not it ever occurs to anyone to ask how they got there. In particular, one can know that there are these ends, goals, or purposes in nature whether or not it ever occurs to anyone to consider the purposes, or even the existence, of a *designer* of nature.

Still, even if (as Aristotle and Aquinas would hold) the existence of such final causes is obvious and unavoidable, it is very odd that there should be such things, and their existence requires explanation even though that explanation, whatever it is, is not something we need worry about for the purposes of everyday scientific research. One of the common objections to the very idea of final causation is that it seems to entail that a thing can produce an effect even before that thing exists. Hence to say that an oak tree is the final cause of an acorn seems to entail that the oak tree – which doesn't exist yet – in some sense causes the acorn to pass through every stage it must reach on the way to becoming an oak, since the oak is the "goal" or natural end of the acorn. But how can this be?

Consider those cases where goal-directedness is associated with consciousness, as it is in us. A builder builds a house, and he is able to do so because the effect, the house, exists as an idea in his intellect before it exists in reality. That is the way in which the house serves as the *final* cause of the actions of the builder even as those actions are the *efficient* cause of the house. Indeed, that is the *only* way the house can do so. For a cause, to have any efficacy, must

in some sense exist; and if it doesn't exist in reality, then the only place left for it to exist (certainly for Aquinas, who as an Aristotelian does not accept Plato's notion of a "third realm" beyond the natural world and the mind) is in the intellect.

What then of the vast system of causes that constitutes the physical universe? Every one of them is directed towards a certain end or final cause. Yet almost none of them is associated with any thought, consciousness, or intellect at all; and even animals and human beings, which are conscious, are comprised in whole or in part of unconscious and unintelligent material components which themselves manifest final causality. But given what was said above, it is impossible for anything to be directed towards an end unless that end exists in an intellect which directs the thing in question towards it. It follows that the system of ends or final causes that make up the physical universe can only exist at all if there is a Supreme Intelligence or intellect outside the universe which directs things towards their ends. Moreover, this intellect must exist *here and now*, and not merely at some beginning point in the past, because causes are *here and now*, and at any point at which they exist at all, directed towards certain ends (otherwise, for reasons examined already, they wouldn't on Aquinas's analysis be true efficient causes at all). As with Aquinas's other arguments, he is not concerned here with whether and how the universe might have begun, but rather with what keeps it as it is at any given moment, a question which must arise even if the universe had no beginning. Hence the Supreme Intelligence of the Fifth Way is not the deistic god that seems to be the most Paley can argue for. Moreover, given his metaphysical assumptions, Aquinas's conclusion follows necessarily and not merely with probability. In these respects (and not only in these respects, as we shall see) the Fifth Way reaches a much stronger conclusion than the design argument, and does so precisely because unlike the design argument it starts from the recognition of the existence of immanent teleology.

Now we saw in chapter 2 that Aquinas regards the final cause as the "cause of causes" insofar as it determines the other causes. In particular, for a thing to have a certain final cause entails that it also has a certain formal and material cause and thus a certain nature or essence; otherwise its final cause would not be inherent in it, nor would it be capable of realizing it. For "upon the form follows an inclination to the end … for everything, in so far as it is in act, acts and tends towards that which is in accordance with its form" (*ST* I.5.5; cf. *QDV* 25.1). But we have also seen that on Aquinas's view, for a contingent thing to be real, its essence must be conjoined to an act of existence, that this can only be accomplished by something outside it, and that the ultimate cause of its

existence must be something in which essence and existence are identical. It follows that whatever orders things to their ends must also be the cause of those things and thus (given what was said earlier) Pure Act or Being Itself. Furthermore, as Garrigou-Lagrange points out, if the Supreme Intelligence were not Pure Act or Being Itself, then its essence would be distinct from its existence, and thus it would have a potency or potential (for existence) which, like all potencies, is of its nature directed towards an end. But in that case there would have to be a higher intelligence directing that potency to its end, and we would be off on exactly the sort of regress that, for reasons we have already seen, must in Aquinas's view terminate in a first member. To explain the reality of final causes, then, we are, once again, unavoidably led to a Supreme Intelligence which is also Pure Act or Being Itself.

Obviously all of this goes beyond what Aquinas says in the text of the Fifth Way itself, which, like the other ways, is intended only as a summary. The point is that, when fully worked out in light of Aquinas's more general metaphysical commitments, the Fifth Way can be seen to lead to precisely the same sort of being whose existence is argued for in the other four ways, by yet another route. And since this being would (if Aquinas's more general metaphysical assumptions are correct) have to be Pure Act or Being Itself, we can see yet again that, if it succeeds, the Fifth Way establishes far more than the finite sort of being reached via the arguments of Paley and his successors in the "Intelligent Design" movement.

Though its approach is very different from Paley's, it might still seem as if the Fifth Way conflicts with Aristotle's view that final causality can exist even in the absence of consciousness. But that there is no conflict here can perhaps be seen by considering the analogy of language. If we consider the words and sentences we speak and write, it is obvious that they get their meaning from the community of language users that produces them, and ultimately from the ideas expressed by those language users in using them. Apart from these users, these linguistic items would be nothing more than meaningless noises or splotches of ink. Still, once produced, they take on a kind of life of their own. Words, sentences, and the like printed in books or recorded on tape retain their meaning even when no one is thinking about them; indeed, even if the books or tapes in question sit in a dusty corner of a library or archive somewhere, ignored for decades and completely forgotten, they still retain their meaning for all that. Moreover, language has a structure that most language users are unaware of, but which can be studied by linguists. And so forth. Still, if the community of language users were to disappear entirely – every single one of them killed in a worldwide plague, say – then the recorded words that were left behind *would* in

that case revert to meaningless sounds or marks. While the community of language users exists, its general background presence is all that is required for meaning to persist in the physical sounds and markings, even if some of those sounds and markings are not the subject of anyone's attention at a particular moment. But if the community goes away altogether, the meaning goes with it. By analogy (and it is only an analogy, and admittedly not an exact one) we might think of the relationship of the Supreme Intelligence of the Fifth Way to the system of final causes in the world as somewhat like the relationship of language users to language. The Supreme Intelligence directs things to their ends, but the system thereby created has a kind of independence insofar as it can be studied without reference to the Supreme Intelligence himself, just as linguists can study the structure of language without paying attention to the intentions of this or that language user. The ends are in a sense just "there" in unconscious causes like the meaning is just "there" in words once they have been written. At the same time, if the Supreme Intelligence were to cease directing things towards their ends, final causes would immediately disappear, just as the meaning of words would disappear if all language users disappeared.

The divine attributes

As we have said, at least when the proofs are read in light of Aquinas's general metaphysical commitments, each of the Five Ways can be seen, if successful, to demonstrate the existence of a being who is Pure Act or Being Itself. Does this mean that they all converge on *one and the same* being, or might the existence of five distinct "gods" be proved via Aquinas's arguments? Aquinas's answer should be clear from what was said in chapter 2 about his doctrine of essence and existence. For the reasons we examined then, on Aquinas's view there can in principle be only one being whose essence and existence are identical, and thus which is Pure Being. Hence it is necessarily one and the same being on which all five proofs converge. This would obviously entail, for the same reason, that there is and can be only one God. For there to be more than one God, there would have to be some essence that the distinct "Gods" all share, each with his own individual act of existence. But since God is that being in whom essence and existence are identical, who just *is* existence or being itself, there is no sense to be made of the idea that he shares an essence with anything else, or has one act of existing alongside others (*ST* I.11.3).

Aquinas also gives two other reasons for holding that the being whose existence is argued for in the Five Ways is necessarily unique. For there to be

more than one such being, there would have to be some way to distinguish one from another, and this could only be in terms of some perfection or privation that one has but the other lacks. But as Pure Act, such a being would be devoid of all imperfections and privations, since imperfections and privations are just different ways in which something could fail to be in act or actual. Hence there can be no way even in principle to distinguish one such being from another, and thus there could not possibly be more than one (*ST* I.11.3). Furthermore, the order that characterizes the world gives it a unity that is explicable only if there is also unity in its cause (*ST* I.11.3).

The *unity* or oneness of God is only one of many divine attributes that Aquinas thinks can be established via pure reason without recourse to divine revelation. We have space here only to provide a brief survey, but Aquinas himself pursues the matter at great length and by deploying a wealth of arguments both in the *Summa Theologiae* and elsewhere, thus exposing as a kind of urban legend the commonly made allegation that even if one were to accept the existence of a first cause, unmoved mover, and so on, Aquinas does nothing to show that such a being would have the other characteristics traditionally ascribed to God. The Five Ways are meant by themselves only to establish the existence of a being having certain key attributes, such as being an unmoved mover, Pure Act, Being Itself, and so forth. Aquinas's next move is to argue that anything having these key features can be seen on analysis necessarily to possess also the other attributes commonly ascribed to God. He follows Pseudo-Dionysius in taking a threefold approach to knowledge of God's attributes (*ST* I.13.8): the way of causality (*via causalitatis*), whereby we move from knowledge of the world to knowledge of God as cause of the world; the way of negation (*via negativa*), whereby we deny of God any characteristic incompatible with his being the first cause and thus Pure Act; and the way of eminence (*via eminentia*), whereby we conclude, by applying the principle of proportionate causality described in chapter 2, that God can be said to possess in an eminent way certain features we attribute to things in the world. As this indicates, while it is sometimes claimed that Aquinas agreed with thinkers like Moses Maimonides that our knowledge of God is purely negative, knowledge of what God is not rather than what he is, this was not in fact his view; indeed, he explicitly repudiates it (*ST* I.13.2). The *via negativa* obviously gives us only negative knowledge of God, but the *via causalitatis* and the *via eminentia* give us some positive knowledge too.

Several attributes seem to follow immediately and obviously from God's being Pure Act. Since to change is to be reduced from potency to act, that which is Pure Act, devoid of all potency, must be *immutable* or incapable of change (*ST*

I.9.1). Since material things are of their nature compounds of act and potency, that which is Pure Act must be *immaterial* and thus *incorporeal* or without any sort of body (*ST* I.3.1–2). Since such a being is immutable and time (as Aquinas argues) cannot exist apart from change, that which is Pure Act must also be *eternal*, outside time altogether, without beginning or end (*ST* I.10.1–2).

As the cause of the world, God obviously has *power*, for "all operation proceeds from power" (*QDP* 1.1; cf. *ST* I.25.1). Moreover, "the more actual a thing is the more it abounds in active power," so that as Pure Act, God must be infinite in power (*QDP* 1.2; cf. *ST* I.25.2). In line with the mainstream classical theistic tradition, Aquinas holds that since there is no sense to be made of doing what is intrinsically impossible (e.g. making a round square or something else involving a self-contradiction), to say that God is omnipotent does not entail that he can do such things, but only that he can do whatever is intrinsically possible (*ST* I.25.3).

The Fifth Way, if successful, establishes by itself that God has *intellect*. Furthermore, intelligent beings are distinguished from non-intelligent ones in that the latter, but not the former, possess only their own forms. For an "intelligent being is naturally adapted to have also the form of some other thing; for the idea of the thing known is in the knower" (*ST* I.14.1). That is to say, to understand some thing is for that thing's essence to exist in some sense in one's own intellect. Now the reason non-intelligent things lack this ability to have the form of another thing is that they are wholly material, and material things can only possess one form at a time, as it were. Hence immaterial beings can possess the forms of other things precisely because they are immaterial; and the further a thing is from materiality, the more powerful its intellect is bound to be. Thus human beings, which, though they have immaterial intellects are also embodied, are less intelligent than angels, which are incorporeal. "Since therefore God is in the highest degree of immateriality … it follows that He occupies the highest place in knowledge" (*ST* I.14.1). This argument presupposes a number of theses in the philosophy of mind and cannot be evaluated, or even properly understood, unless those theses are first understood. We will explore these theses in chapter 4.

We can also conclude, in Aquinas's view, that "there is will in God, as there is intellect: since will follows upon intellect" (*ST* I.19.1). Why do will and intellect necessarily go together? For Aquinas, things naturally are inclined or tend towards their natural forms, and will not of themselves rest, as it were, until that form is perfectly realized; hence the acorn, for example, has a built-in tendency towards realizing the form of an oak, and will naturally realize that form unless somehow prevented by something outside it. What we are describing in this

example is of course the goal-directedness of the acorn as something having a final cause. But other sorts of thing have final causes too. In sentient beings, namely animals, this inclination towards the perfection of their forms is what we call appetite. And in beings with intellect it is what we call will. Thus anything having an intellect must have will. (We will return to this topic in the next chapter.) Of course, since God does not have the limitations we have, he does not have any ends he needs to fulfill, any more than he needs to acquire any knowledge. Thus, as with our attribution of power, intellect, and other attributes to God, our attribution of will to him is intended in an *analogous* rather than a univocal sense.

Since something is perfect to the degree it is in act or actual, God as Pure Act must be *perfect* (*ST* I.4.1). Given the convertibility of being and goodness, God as Pure Act and Being Itself must also be *good*, indeed the highest good (*ST* I.6). At this point, it might be objected that the problem of evil casts doubt on this claim; for if God is good, why hasn't he eliminated the evil that obviously exists in the world? But there are several problems with this objection. First of all, it could only undermine Aquinas's argument for God's goodness if we assumed that a good being could not possibly have a reason to allow evil. But it is notoriously difficult to show that such a being *could not possibly* have such a reason, and even most contemporary atheist philosophers would not make such a strong claim. In the absence of such an assumption, though, Aquinas could simply insist that since his arguments have proven that God exists and is good, it follows that whatever evil exists must be consistent with his goodness.

Second, as Aquinas himself argues in reply to the problem of evil, "this is part of the infinite goodness of God, that He should allow evil to exist, and out of it produce good" (*ST* I.2.3). That is to say, the reason God allows evil is precisely because he intends to bring good out of it, whether that good is that we come to learn from the mistakes we make, that we come to have certain virtues that could not be acquired without struggling against evil, that we come to appreciate what is good by contrast with what is evil, or whatever. And given that God is omnipotent and that (as we shall see in the next chapter) Aquinas holds that we have immortal souls, so that our time on earth is merely a brief prelude to an everlasting existence in the hereafter, it is hardly implausible to suggest that God is capable of rewarding us with a good in the next life that is so tremendous that even the most horrendous evils in this life will come to seem trivial in comparison, and worth having suffered through. Of course, in the face of the worst real-world evils, the idea of such a good in the hereafter can seem cold, abstract, and remote. But that is an emotional problem rather than an intellectual one; it has no tendency to show that there is or could be no such good, but only

that it is hard for us to keep our minds fixed on it in the face of suffering. Nor could an atheist dismiss such a response out of hand without begging the question. To say "There is no God, because of all the unredeemable evil that exists; and the evil that exists must be unredeemable, because there is no God who could redeem it," would be to argue in a circle.

Third, as Brian Davies has emphasized, much discussion of the problem of evil seems to presuppose that God is a kind of moral agent who has certain duties which (so it is alleged) he has failed to live up to. But this way of thinking simply makes no sense given Aquinas's conception of God. For only creatures with the sorts of limitations we have can coherently be described as having moral duties. For example, given that we depend on other people for our well-being and they depend on us, we have certain obligations towards each other; given that we have certain potentials the realization of which is good for us, potentials which require a certain amount of effort to realize, we have a duty to make that effort; and so forth. But as Pure Act and Being Itself, God has none of these dependencies, potentials, or limitations, and thus there is no sense to be made of the suggestion that he either has or lacks this or that moral virtue or has lived up to or failed to live up to this or that moral obligation. Though his possession of intellect and will (or, more precisely, of something analogous to what we call intellect and will in us) entails that he is in some sense "personal" (rather than the sort of impersonal deity familiar from certain Eastern religions), God is nevertheless not "a person" in the sense that we are, with all the limitations that expression implies.

That God is very remote indeed from the things of our experience is nowhere clearer than in Aquinas's account of divine *simplicity*, which is perhaps the most controversial aspect of his teaching on the divine attributes. For Aquinas, God is "simple" in the sense of being in no way composed of parts (*ST* I.3). As has been said, he is incorporeal and immaterial, and thus cannot have any bodily parts nor be composed of form and matter. But neither does he have even any metaphysical parts. For as we have also seen, on Aquinas's account there is no distinction between essence and existence in God. Unlike everything else that exists, he just *is* his own existence, and just *is* his own essence, for these are identical. For this reason, there can also be no distinction between genus and difference in God, since *being*, the only candidate genus for something whose essence and existence are identical is (as we saw in chapter 2) no genus at all, and since for there to be a member of a genus, it must have an act of existence which differs from the essence it shares (at least potentially) with other members of the genus, and, again, there is no distinction between essence and existence in God. Hence, again, "it is clear that God is nowise composite, but is altogether

simple" (*ST* I.3.7).

One famous implication of this doctrine is that though we distinguish in thought between God's eternity, power, goodness, intellect, will, and so forth, in God himself there is no distinction between any of the divine attributes. God's eternity *is* his power, which *is* his goodness, which *is* his intellect, which *is* his will, and so on. Indeed, *God himself* just is his power, his goodness, and so on, just as he just *is* his existence, and just *is* his essence. Talking or conceiving of God, God's essence, God's existence, God's power, God's goodness, and so forth are really all just different ways of talking or conceiving of one and the very same thing. Though we distinguish between them in thought, there is no distinction at all between them in reality. For, again, if there were such a distinction, then we could distinguish parts in God, and being absolutely simple, God has no parts.

Though the idea of divine simplicity might seem odd or eccentric to some contemporary readers, it is historically speaking the mainstream view of God's nature within the classical theistic tradition, being defended not only by Aquinas, but by thinkers as diverse as St. Athanasius, St. Augustine, St. Anselm, Maimonides, Avicenna, and Averroes, to name just a few. It is affirmed in such councils of the Roman Catholic Church as the Fourth Lateran Council (1215) and Vatican I (1869–1870). Nevertheless, it has been criticized by a number of contemporary philosophers and theologians. Some of this criticism derives from objections to Aquinas's doctrine of the identity of essence and existence in God, which were discussed in chapter 2. Some of it derives from worries over whether it makes sense to say that God's power is identical to his goodness, which is identical to his intellect, and so on; for wouldn't this entail that power is the same thing as goodness, goodness the same thing as intellect, and so forth, which is obviously false? But this does not in fact follow. For one thing, just as (to use Frege's famous example) we can acknowledge that the expressions "the morning star" and "the evening star" differ in *sense* while consistently affirming that they *refer* to one and the same thing (the planet Venus), so too can we acknowledge the obvious fact that "power," "goodness," "intellect," and so on differ in sense while insisting that when applied to God they refer to one and the same thing. For another thing, we must keep in mind Aquinas's doctrine of analogy, according to which, while the terms we apply to created things do not apply to God in either equivocal or univocal senses, they do apply in analogical senses. So, while it would of course be absurd to say that power, goodness, intellect, and so forth are all identical in God if we were using these terms in exactly the same sense in which we apply them to ourselves, it is not absurd to say that there is in God something that is *analogous* to power, something *analogous* to goodness,

something *analogous* to intellect, and so on, and that these "somethings" all turn out to be one and the same thing.

As Eleonore Stump has noted, there are also in any case certain advantages of the doctrine of divine simplicity that ought to recommend it at least to philosophers otherwise sympathetic with Aquinas's attempt to prove the existence of God via cosmological arguments and/or with the idea that there is an important link between the existence of God and the foundations of morality. To take the latter issue first, there is a distinction commonly drawn between the view that something counts as good or bad because of God's will and the view that God wills for us to do or to avoid doing something because it is either good or bad by reference to some standard external to him, and neither of these views is theologically satisfactory. The first seems to make morality entirely arbitrary, insofar as it appears to entail for example that torturing infants for fun would have been good if God had willed this. The second seems to entail that morality is ultimately independent of God, which seems incompatible with the idea that everything that exists other than God ultimately derives from him. This is the basis for the "Euthyphro dilemma" (named after a dialogue of Plato's in which it was famously put forward), which is an attempt to refute the thesis that morality depends on God by arguing that there are only these two ways of understanding this thesis and that neither of them can be acceptable to the theist. But as Stump points out, the doctrine of divine simplicity shows that there is a third option here, so that the dilemma in question is a false one. For if God just *is* perfect goodness which just *is* the divine will which just *is* immutable and necessary being, then there can be no question either of God willing in accordance with some standard of goodness independent of him or of his will being arbitrary. What is objectively good and what God wills for us as morally obligatory are just *the same thing* considered under different descriptions, and neither could have been other than they are. (We will return to this issue when we examine Aquinas's moral theory in chapter 5.)

Stump notes also that the doctrine of divine simplicity affords certain advantages to versions of the cosmological argument informed by it. For example, as we noted earlier, some modern versions of the cosmological argument hold that only God can serve as the ultimate explanation of why the universe exists because he is a being whose essence "includes" existence, as if existence were one "property" of God alongside and distinct from other ones, a property which is also distinct from the essence which "includes" it and from the divine being who "possesses" it. The trouble with this is that it seems perfectly possible to detach "existence" so conceived from God's other "properties"; certainly the assertion that existence must necessarily go together with them

seems arbitrary and itself in need of explanation. But if God just *is* his existence which just *is* his power, which just *is* his will, and so on, then this problem disappears.

Much more could be said about Aquinas's account of the divine attributes, but this much suffices to show that there is no basis whatsoever for the widespread assumption that Aquinas never justifies the claim that the being whose existence he argues for in the Five Ways is the God of traditional theism. It also gives a sense of how much Aquinas thinks we can know about God through purely philosophical reasoning. But there is also a sense in which Aquinas thinks that we ultimately cannot know the *essence* of God, at least not as it is in itself. For in the strict sense knowledge of the essence of a thing requires the ability to define it in terms of its genus and difference, and as we have seen, there is for Aquinas no distinction in God between genus and difference, and thus no way to define him (*CT* 26). It is in *this* sense that Aquinas holds that "we cannot know what God is, but rather what He is not" (*ST* I.3). And this is why the famous ontological argument associated with St. Anselm is not considered by Aquinas to be one of the ways in which we might prove the existence of God. For Anselm, God is by definition the greatest conceivable being, and it is (Anselm holds) greater to exist than not to exist. Hence if God did not exist it would follow, absurdly, that there could be something conceivably greater than the greatest being. Anselm's argument thus begins with a definition of God's essence and attempts to show that given knowledge of that essence, we can know also that there must be something in reality corresponding to it, and thus that God exists. Since Aquinas holds that God's essence and existence are identical, he agrees that if we could have knowledge of God's essence we could see that he must exist. But since in fact we cannot, in his view, have knowledge of that essence, we cannot know the starting point of the ontological argument (*ST* I.2.1). Our knowledge of God must therefore be *a posteriori*, based on observation of his effects. But that, as we have seen, affords us in Aquinas's view with ample grounds indeed for affirming God's existence and predicating of him the traditional divine attributes.